Tacinf: A Unified Tactile Infrastructure for Robot Manipulation

Qiwei Wu[12*‡] **Xiangyu Chen**[2*] **Xin Xiang**[2*] **Rui Zhang**[1] **Yihang Kang**[2] **Weiran Chen**[2] **Zikang Xie**[2] **Yangfan Li**[2] **Tao Wang**[2] **Xiangzheng Wang**[2] **Tianyi Bi**[2] **Zhiyan Zhang**[2] **Xianpeng Wang**[2] **Mingqian Zhou**[3] **Yan Yu**[2] **Chuanyu Li**[2] **Wenjie Zhang**[2] **Tao Li**[1] **Weihua Zhang**[1†] **Renjing Xu**[2†] **Junjie Lai**[1]

[1]Nvidia [2]HKUST(GZ) [3]University of Cambridge

[*]Equal Contribution [‡] Project Leader [†] Corresponding Authors

tacinf.github.io

Abstract

Force-aware manipulation requires a robot to infer appropriate interaction forces from task context and tactile feedback. We present Tacinf, a low-cost modular infrastructure for this setting. It enables policies to predict robot actions and fingertip-force outputs from task language, vision, and touch; an optional language-conditioned force level can be provided when explicit force regulation is required, while ordinary task instructions need no force adverbs. Tacinf integrates simulation data synthesis and a tactile benchmark (Tabero), policy learning (VTLA), handheld data collection with a shared deployment gripper (Tacumi), and task-aligned simulation-to-real evaluation (Taco), forming a full-stack closed loop from multi-source mixed-data collection through the foundation model to the hardware system. We validate the effectiveness of each component, including the simulation data, simulation benchmark, controller, and Tacumi, and are open-sourcing the full stack. We achieve zero-shot transfer of a foundation model fine-tuned with tactile simulation data only to real-robot tasks, and show across multiple policy variants that simulation preserves the real-robot success ordering well (Pearson $r = 0.998$, MMRV$= 0.010$), while the force component exhibits a small sim-to-real gap. Co-training experiments show that the mixed-training scheme over teleoperation, simulation, and Tacumi data balances collection time and data utility: with less data, it achieves force-aware instruction following on position-randomized out-of-distribution objects, raising the success rate from 20% to 60%.

Keywords: Tactile Sensing, Vision-Language-Action Models, Language-Conditioned Force Control, Zero-Shot Sim-to-Real Evaluation

1. Introduction

Robot manipulation foundation models have made rapid progress, but force-aware tasks require the policy to do more than select a motion from images and task language: it must also infer and regulate physical interaction. This includes not only tasks with explicit interaction instructions, such as operating “gently” or “firmly,” but also ordinary task instructions whose force requirement is implicit, such as opening a container or wiping a surface, so the policy must infer suitable contact behavior from the instruction, scene, and tactile feedback. Tactile data are essential for training a suitable manipulation foundation model in these settings, yet heterogeneous hardware and sensor designs make building effective datasets costly. Even with a suitable model trained from data, the system must adjust quickly and effectively through real-time feedback and apply appropriate force to the target. When a policy performs poorly, it can damage the object or the robot itself, so a low-cost evaluation platform for testing and iterating on policies is also important.

To address these needs, we introduce Tacinf (Figure 1), a system-level, full-stack open-source infrastructure

© 2026 NVIDIA. All rights reserved.



Figure 1: **Tacinf combines tactile data collection, force-aware evaluation, and sim-to-real deployment.** The left panel shows Quest 3 teleoperation, UMI, and a force-controlled gripper. The center panel presents LEGO-like VTLA, scalable tactile tasks, and a force-aware evaluation protocol. The right panel shows zero-shot Sim2Real evaluation and co-training with UMI and simulation-augmented data.

designed for tactile and force-aware tasks. Tacinf-Tabero provides a simulation-based way to obtain tactile and force-supervised data together with a method for evaluating VTLA policies; Tacinf-VTLA provides a modular tactile interface for rapidly instantiating different tactile policies; Tacinf-Tacumi provides hardware for applying policies to physical robot systems and for collecting teleoperation and handheld tactile/force-supervised data; and Tacinf-Taco performs task-aligned simulation-to-real evaluation and supports multi-source data mixing. Together, these components provide a framework for multi-source data collection, simulation evaluation and sim-to-real transfer, and policy deployment for force-aware tasks, at low hardware, compute, and data cost; the Tacumi kit costs under $2,500, and the minimum compute requirement is a single RTX 4090.

In our experiments, we validate the effectiveness of each system component and obtain the following key findings: (1) after fine-tuning a tactile foundation model with simulation data only, we achieve zero-shot deployment on real-robot tasks, maintaining a success rate of 98% or above on LIBERO-like tasks in the real-robot setting, with force prediction reaching human-level performance; (2) across eight policy variants and six tasks designed with Tacinf-VTLA, simulation preserves the real-robot success ordering well (Pearson $r = 0.998$, MMRV$= 0.010$) with a small measured-force sim-to-real gap; and (3) teleoperation, Tacumi, and simulation data combine effectively: we verify the complementary role of each source, including the use of data collected without a robot embodiment, and at lower data cost raise the success rate from 20% to 60% at randomized task positions and on out-of-distribution objects.

Our contributions are:

- **Multi-source tactile data collection and synergy:** we build a low-cost, open-source tactile data pyramid and a multi-method, multimodal data-collection system, and validate the synergistic effects of its sources through ablation studies.
- **Force-following and execution system:** using multi-source data, we build a complete system from model to actuator that enables scalable force control on contact-rich tasks.
- **Simulation consistent with the real robot:** we propose an alignment procedure between the tactile manipulation foundation-model simulator and the physical robot, achieving highly consistent behavior while reducing evaluation cost.

2. Related Work

2.1. Simulation Platforms and Synthetic Data

Robot simulation provides reproducible environments for policy learning and evaluation. Early infrastructures such as robosuite, LIBERO, and CALVIN established reusable manipulation tasks and language-conditioned evaluation protocols Liu et al. (2023); Mees et al. (2022); Zhu et al. (2020). Recent platforms extend these capabilities through household scene diversity in RoboCasa, GPU-parallel simulation in ManiSkill3 and Isaac Lab, and unified assets and interfaces in RoboVerse Geng et al. (2025); Nasiriany et al. (2024); NVIDIA et al. (2025b); Tao et al. (2025). MuBlE couples MuJoCo dynamics with Blender rendering for tasks requiring both visual and physical reasoning Nazarczuk et al. (2025). At the data-generation level, RoboTwin and RoboTwin 2.0 combine task construction and demonstration generation for bimanual manipulation, while SkillMimicGen reuses demonstrations through skill-level generation Chen et al. (2025); Garrett et al. (2025); Mu et al. (2025). GraspVLA further demonstrates the use of large-scale synthetic action data for policy pretraining Deng et al. (2025). These approaches motivate our reuse of existing trajectories, but trajectory replay alone does not establish that contact forces or tactile observations transfer across simulators.

A complementary direction reconstructs real environments for simulation-based policy evaluation. SIMPLER studies visual and control gaps and evaluates whether simulated performance predicts real-robot behavior; PolaRiS combines neural scene reconstruction with simulation-data co-training Jain et al. (2026); Li et al. (2025b). SimFoundry automates scene reconstruction and generates task-relevant scene variations, SceneRig reconstructs simulation-ready 3D scenes from single images Luo et al. (2026); Ranawaka et al. (2026), while Seed3D 2.0 provides simulation-oriented 3D asset generation Gu et al. (2026). Ctrl-World instead uses an action-conditioned generative world model for policy evaluation and improvement Guo et al. (2026). These works establish precedents for real-to-sim reconstruction, data mixing, and cross-domain policy ranking. Tacinf-Taco adapts this direction to tactile observations and force-conditioned evaluation.

Tactile simulation datasets and benchmarks are also emerging. VTDexManip supports visual–tactile pretraining and dexterous manipulation, SafeDiff evaluates force-aware interaction using SafeDoorManip50k, and UniVTAC unifies multi-sensor tactile synthesis, representation learning, and manipulation benchmarking Chen et al. (2026); Liu et al. (2025b); Wei et al. (2025). SoftVTBench pairs visuotactile observations with independent deformation ground truth for evaluating deformable-object interaction quality Jing et al. (2026). OpenViTac benchmarks visuo-tactile policies in paired simulation and real-world settings and reports sim-to-real agreement across policy families Wu et al. (2026). Tacinf-Tabero focuses on converting existing manipulation trajectories into synchronized vision–tactile–language supervision and on evaluating instruction-dependent interaction forces alongside task completion.

2.2. Tactile Simulation Methods

Vision-based tactile simulation must account for contact geometry, sensor deformation, and image formation. TACTO couples physics-engine contact information with efficient optical rendering Wang et al. (2022). Taxim calibrates an example-based optical mapping and models marker motion using elastic deformation and superposition; FOTS provides fast optical and marker-motion approximations for tactile-motor learning Si and Yuan (2022); Zhao et al. (2024). These methods offer efficient sensor-observation models, although their calibration assumptions and approximations must be distinguished from a complete elastomer dynamics model. TacEx integrates soft-body contact simulation with Taxim and FOTS in Isaac Sim, and TacSL provides a GPU-parallel visuotactile library in which optical rendering and marker-field extraction are driven by the same simulated deformation, so that the image and marker signals stay congruent, together with policy-learning support on those observations Akinola et al. (2025); Nguyen et al. (2024).

Physics-based approaches pursue more explicit contact mechanics. DiffTactile combines differentiable sensor

and object simulation for system identification and trajectory optimization, while Taccel integrates contact and rigid-body dynamics in a GPU-parallel tactile simulation platform Li et al. (2025c); Si et al. (2024). HydroShear additionally models path-dependent shear accumulation and stick–slip transitions Dang et al. (2026). Learned observation transfer offers a complementary route: SimShear generates shear-aware tactile images from simulated observations and shear conditioning, and Cross-Sensor Touch Generation translates tactile observations between sensor designs Freud et al. (2025); Rodriguez et al. (2025). Applications such as AnyRotate, UniTac2Pose, and VT-Refine demonstrate how simulated touch can support in-hand manipulation, pose estimation, and policy refinement Huang et al. (2025b); Wu et al. (2025b); Yang et al. (2025). Together, these studies expose a tradeoff between contact fidelity, calibration effort, and simulation throughput, and they establish tactile sim-to-real transfer as an important evaluation target Freud et al. (2025); Yang et al. (2025). Following the modular integration exemplified by TacEx, Tacinf uses Taxim for optical observations and FOTS for marker motion; our contribution concerns their calibrated integration into a scalable data and evaluation pipeline and the evaluation of language-conditioned force-level transfer across multiple policy variants, rather than a new sensor-mechanics solver or a reduction of the tactile sim-to-real gap. Compared with a simulator that generates both streams from one simulated deformation, such as TacSL, this modular choice trades per-sample deformation congruence for the throughput needed to replay open-source trajectories at scale; the marker stream is therefore used as a relative shear and direction cue, while the wrench used for supervision and control is taken from the physics-engine contact reactions rather than decoded from the tactile image.

2.3. Tactile Data Collection

Learning generalizable robot manipulation policies requires diverse, high-quality demonstrations, yet collecting real-world robot interaction data remains costly and difficult to scale. Teleoperation is a widely used approach in which human operators guide physical robots to perform tasks, providing demonstrations that directly reflect the robot's observations and actions. Large-scale efforts such as DROID demonstrate the value of diverse teleoperated data Khazatsky et al. (2024). However, each collection setup requires access to a robot, and scaling collection entails additional hardware, operator time, and deployment effort. Universal Manipulation Interface (UMI) addresses these constraints by collecting demonstrations with handheld grippers and transferring the resulting skills to robots, decoupling demonstration collection from physical robot access Chi et al. (2024). FastUMI further develops this direction through a scalable, hardware-independent collection interface Zhaxizhuoma et al. (2025). For tactile learning, sensor hardware and observation alignment are equally important: GelSight provides high-resolution contact geometry and deformation measurements, PolyTouch combines complementary sensing modalities, and 3D-ViTac fuses distributed tactile measurements with visual geometry Huang et al. (2025a); Yuan et al. (2017); Zhao et al. (2025a). These approaches provide a practical foundation for expanding manipulation datasets beyond dedicated robot workstations.

For contact-rich manipulation, demonstrations must capture not only motion and visual observations but also the contact information needed to regulate physical interaction. In teleoperation systems without real-time force feedback, operators rely primarily on visual observations to infer contact states, making it difficult to perceive and regulate interaction forces accurately. Insufficient or excessive force may result in unstable grasps, excessive object loading, or task failures. These issues can compromise demonstration quality and increase repeated attempts and unusable demonstrations, reducing collection efficiency. Moreover, collecting tactile demonstrations through teleoperation still requires sensor-equipped robots, limiting scalability. Integrating tactile sensors directly into handheld manipulation interfaces such as UMI allows operators to manipulate objects directly while synchronously recording motion, vision, and touch. Combined with direct mechanical feedback or active motor feedback, these interfaces allow operators to perceive contact states and adjust interaction forces more intuitively. ViTaMIn combines a compliant handheld gripper with visuotactile sensing and operator-perceptible force feedback, while ViTaMIn-B extends this approach to bimanual collection with

tracked controllers Li et al. (2025a); Liu et al. (2025a). FreeTacMan uses a wearable gripper for synchronized visual–tactile demonstrations, exUMI integrates modular tactile sensing with calibrated state acquisition, and TacUMI combines tactile observations, force–torque measurements, and pose tracking Cheng et al. (2026a); Wu et al. (2025a); Xu et al. (2025b). UMI-FT further demonstrates the use of finger-level wrench measurements for learning compliant manipulation Choi et al. (2026). Building on this direction, Tacinf-Tacumi combines tactile sensing with real-time active force feedback, enabling operators to perceive and regulate grasp effort during demonstration collection. The handheld interface and robot execution system use the same low-cost gripper motor, eliminating the need to select a separate, dedicated gripper motor for deployment. This reduces additional hardware procurement and adaptation effort, lowering deployment costs and integration complexity.

2.4. Tactile Foundation Models

Tactile foundation models encompass transferable perception backbones as well as policies conditioned on touch. Learning to Jointly Understand Visual and Tactile Signals and UniTouch align tactile observations with visual or multimodal representations Li et al. (2024); Yang et al. (2024). Sparsh studies self-supervised tactile pretraining and introduces TacBench; Transferable Tactile Transformers uses a shared backbone with sensor-specific and task-specific modules Higuera et al. (2025a); Zhao et al. (2025b). Sensor-Invariant Tactile Representation and AnyTouch address cross-sensor transfer, with AnyTouch explicitly combining static and dynamic tactile information Feng et al. (2025); Gupta et al. (2025). TacX extends pretraining beyond tactile images to multiple sensing modalities, while AnyTouch 2 emphasizes temporal deformation and force-aware dynamics Feng et al. (2026); Higuera et al. (2025b). These advances establish that scalable touch learning depends on sensor diversity, temporal information, and appropriate transfer evaluations, not simply on treating a tactile image as another camera view.

Tactile–language models connect contact observations to semantic reasoning. Octopi and Octopi-1.5 use tactile inputs for object-property inference and language-grounded reasoning Yu et al. (2024, 2025b). On the action side, OpenVLA, OpenVLA-OFT, π_0, $\pi_{0.5}$, RDT-1B, and GR00T N1 provide generalist policy architectures and training recipes that motivate tactile extensions Black et al. (2025a,b); Kim et al. (2025a,b); Liu et al. (2025c); NVIDIA et al. (2025a). TLA and VTLA study language-conditioned contact-rich manipulation, including insertion and preference-based learning Hao et al. (2026); Zhang et al. (2026). TA-VLA examines torque-aware policy design; OmniVTLA aligns heterogeneous tactile signals with semantic representations; VLA-Touch introduces tactile feedback at planning and execution levels Bi et al. (2026); Cheng et al. (2026b); Zhang et al. (2025b). Tactile-VLA already combines tactile-conditioned prediction with hybrid position–force control Huang et al. (2025c). Related visuotactile policies such as RoboPack and Reactive Diffusion Policy use touch for dynamics inference or fast reactive control Ai et al. (2024); Xue et al. (2025).

Tacinf-VTLA builds on these representation and control directions as a simple, modular interface framework that enables tactile and force information to be applied during foundation-model post-training. It studies alternative tactile inputs, predicts fingertip-force targets, and couples a pretrained policy to separate grip-force and applied-wrench feedback loops.

3. Tacinf

3.1. Tacinf-Tabero

Tacinf-Tabero is the simulation benchmark component of Tacinf. It provides reproducible language-conditioned tactile manipulation tasks, scalable multimodal data generation, and process-aware metrics for evaluating physical interaction quality.

3.1.1. Data Reuse

Open-source robotic manipulation datasets are valuable community resources. To avoid rebuilding every dataset from scratch, we design a reusable workflow that transcribes datasets originally developed in MuJoCo or other platforms into the Isaac Sim-based Tacinf-Tabero benchmark. The transcription improves visual fidelity through high-quality rendering and augments the data with tactile information.

We first reconstruct task environments in Isaac Lab NVIDIA et al. (2025b) to match the source domains, including assets, initial object and robot poses, and success logic. This workflow can currently be partially automated with SimFoundry Ranawaka et al. (2026) or large language models. We then replay the source trajectories in the reconstructed scenes to obtain multimodal data that follow the demonstrated human trajectories. Direct migration of tactile-aware tasks faces two challenges. First, the end-effector differs: our robot uses a gripper integrated with visuo-tactile sensors whose geometry differs from the original. Second, the underlying physical behavior differs: some source datasets rely on Operational Space Control (OSC) Khatib (1987), which is highly sensitive to physical parameters. Datasets collected by real-robot teleoperation or UMI devices also introduce simulation-to-real gaps. We therefore make two adjustments: (1) align the end-effector tool center point (TCP) by adjusting the robot-arm base pose; and (2) use a high-gain PD joint controller during trajectory replay to minimize cumulative tracking error.

After evaluation, the robotic arm equipped with a tactile gripper achieves an acceptable task success rate during data replay, which we refer to as the data retention rate.

3.1.2. Cross-Modal Data Acquisition

Leveraging the GPU-accelerated parallel rendering capabilities of Isaac Lab, we build a synchronized and scalable multimodal data acquisition system that captures diverse sensory streams with a shared temporal convention. Visual information is obtained from two camera configurations: a wrist-mounted camera and a third-person view camera, both providing synchronized RGB-D data. Tactile observations are produced by a simulated GelSight Yuan et al. (2017) sensor with a resolution of 320×240, providing both RGB tactile images and an 11×9 marker grid. We also support Xense sensors with a marker grid of 20 rows and 11 columns; image resolution is specified by the sensor configuration. To maximize simulation speed, we implement the tactile perception simulation by following the computation schemes of Taxim and FOTS Si and Yuan (2022); Zhao et al. (2024), which are computationally lightweight.

Tactile Image. Tactile images are generated using the Taxim framework Si and Yuan (2022). GelSight builds on retrographic sensing Johnson and Adelson (2009) and reconstructs contact geometry from calibrated illumination-dependent images Yuan et al. (2017); in simulation, the geometry-to-image mapping is evaluated in the forward direction. In simulation, Taxim shifts the depth map using the indentation depth, approximates the gel's static soft-body deformation through multi-layer Gaussian filtering, and then computes surface normals and performs optical rendering. **Marker Displacement Field.** Following the FOTS approach Zhao et al. (2024), we model the displacement field of surface markers on an elastic tactile sensor. Given simulated contact deformation and relative-motion cues, Gaussian spatial decay functions approximate marker displacement caused by normal expansion, shear, and planar torsion. The resulting field is expressed in sensor image coordinates and provides a raw but informative signal for estimating shear forces and torques. This marker-field interface can be extended from optical tactile sensors to capacitive, piezoelectric, or magnetic sensors.

Contact Force. The physics engine provides the left and right sensor reactions, $\mathbf{f}_L^{S_L}$ and $\mathbf{f}_R^{S_R}$, in their local frames. After rotation into the TCP frame, the negative sum gives the force exerted on the object. The two contact forces and their moment arms provide a point-force approximation of the TCP wrench (Section 3.2); the scalar grip proxy is computed separately from the contact force magnitudes.

All cameras are rendered in parallel using tiled rendering, and all modalities, including visual, tactile, force,

language instructions, and executed actions, are sampled synchronously at 20 Hz to produce temporally aligned multimodal observations at each time step.

3.1.3. Language-Conditioned Force-Level Data Generation

We focus on tasks in which the desired interaction cannot be specified adequately by task completion alone, including explicit force instructions and contact-sensitive object handling. We distinguish two categories: (1) language-conditioned force-level tasks; and (2) tasks oriented toward deformable objects or objects with articulation parts. Tabero is primarily used to validate the first category, while we validate the corresponding behavior for the second category directly in real-world scenarios.

For the first category of tasks, most open-source datasets use continuous actions for arm control but binary or discrete gripper commands, limiting fine-grained force regulation and producing narrow within-task force distributions. We address this by executing the same task with different force magnitudes through low-level gripper-controller parameters.

In our simulation, the gripper uses a spring–damper position servo, consistent with the impedance-control formulation Hogan (1985). For a single-DOF gripper with width p, the commanded force is:

$$F^{\text{cmd}} = K_p(p^{\text{target}} - p) + K_d(\dot{p}^{\text{target}} - \dot{p}). \tag{1}$$

At steady state after contact, velocities vanish and the static grip force approximates $F_{\text{static}} \approx K_p \cdot \delta$, where $\delta = p - p^{\text{target}} > 0$ is the penetration depth. Thus, varying K_p directly scales the steady-state force under identical high-level commands. During initial contact, however, the impact force is dominated by damping: $F_{\text{impact}} \propto K_d \cdot |\dot{p}|$, so adjusting K_d diversifies transient forces. This variation is difficult to realize directly with many real robot grippers, motivating the tactile UMI device described below for data collection.

We replay trajectories under different (K_p, K_d) settings to instantiate an ordered force-level condition ℓ. In the reported benchmark, $\ell \in \{10\%, 25\%, 100\%\}$, with the same task and trajectory semantics retained across levels. These interactions produce distinct sensor-side force patterns $\mathbf{f}_L^{S_L}$ and $\mathbf{f}_R^{S_R}$; their TCP-frame transforms and object-side signs follow Eq. (4). Following Tactile-VLA Huang et al. (2025c), we attach language such as “gently” or “softly” to low-force conditions and “firmly” or “tightly” to high-force conditions, aligning the language condition with the measured force distribution. Here “gentle” is one endpoint of a force-level following problem rather than the complete objective: the policy should preserve the ordering and separation induced by ℓ. For special tasks whose nature already demands substantial force, such as wiping a blackboard, we do not add an adverb. We also record continuous gripper aperture p rather than discrete open/close signals, preserving compatibility between the action space and force dynamics. This pipeline provides a reusable way to generate language-conditioned force profiles for arbitrary tasks.

3.1.4. Metrics Beyond Success Rate

Task success alone overlooks critical aspects of physical interaction; force-aware and deformation-aware benchmarks address complementary aspects of this issue Jing et al. (2026); Wei et al. (2025). To reduce simulation compute cost, we adopt a rigid-body plus force-sensing approach to approximate the contact process. We introduce the forces obtained at the gripper fingertips as evaluation metrics to quantify the quality of physical interaction during task execution. The two grip-force metrics use F_{grip}, derived from the squeeze proxy in Eq. (2) using the relation in Eq. (3): **Maximum Transient Grip Force (MG).** The average of the top 5% values of F_{grip} over an episode, capturing peak grasping effort and transient spikes that may indicate aggressive or damaging behavior. **Average Grip Force (AG).** The stable grip force of both fingers, represented by F_{grip} over the process when the object does not slide relative to the fingertips. It serves as a reference for tangential squeeze, rather than an absolute ground-truth value. **Maximum Transient Applied Force (MA).** The average of the top 5% magnitudes of the applied force, characterizing extreme interaction events such as impacts.



Figure 2: **Tacinf-Tabero simulation benchmark.** Tacinf-Tabero replicates the LIBERO task environments, enables data reuse, enhances the visual fidelity of simulated data, and makes it possible to obtain high-quality tactile modalities.

Average Applied Force (AA). The mean applied force magnitude during contact, measuring overall interaction intensity.

With wrench-aware control, the three translational forces and the two modeled torque components about TCP x and z are additionally tracked in terms of their means, peaks, and tracking errors. Under the sufficient-grip, no-slip assumption below, object-dependent torque about the grasp axis is excluded from this sensor-centered evaluation. Together, these metrics enable a more nuanced and physically grounded evaluation of robot policies, moving beyond binary success to assess safety and interaction quality.

3.2. Tacinf-VTLA

Tacinf-VTLA is not a specific model architecture but a modular framework that equips foundation vision–language–action models with tactile sensing and language-conditioned force-level control; this paper instantiates it on the OpenPi infrastructure associated with π_0 and $\pi_{0.5}$ Black et al. (2025a,b) as concrete examples. The instantiated policy component predicts fingertip-force targets from vision, language, and touch and incorporates a closed-loop force–position control strategy that converts its outputs into compliant robot motion.

3.2.1. Fingertip-force Decomposition and Two-contact Wrench Mapping

The separation of internal grasp forces from the resultant object wrench is standard in grasp mechanics; recent compliant imitation-learning systems also distinguish grasp and external interaction forces Choi et al. (2026). Our force decomposition also draws on the force-constrained formulation used in gentle-manipulation policy learning Zhou et al. (2026). In our implementation, a two-finger parallel gripper has a single opening–closing degree of freedom, so the planar contact at its fingertips decomposes naturally into two parts: the components squeezing each other along the closing axis form the *grip force*, while the remaining components constitute the *applied interaction* on the object, represented as an estimated six-dimensional wrench at the TCP under a two-contact approximation (3D force plus a 3D moment vector whose grasp-axis component is neglected). During grasping, the normal force that compresses the sensor pads against the object surface dominates; when the gripper transports, wipes, or twists the object, shear forces and moments equally determine the object motion.

3.2.2. Hybrid Controller

Building on hybrid position–force control and impedance-based interaction control Hogan (1985), as well as their use in tactile-conditioned policies Choi et al. (2026); Huang et al. (2025c), we implement force-error-driven pose and gripper-width corrections. We distinguish a scalar squeeze channel from the applied wrench, which comprises a 3D force and a 3D torque. Let $i \in \{L, R\}$ index the left and right fingertips, S_i denote each sensor frame, and T denote the TCP frame. A superscript S_i or T specifies the coordinate frame, whereas $\top$ denotes transpose. Lowercase $\mathbf{f}_i^{S_i} \in \mathbb{R}^3$ is the contact reaction acting on sensor i; its local z-axis is normal to the pad. The squeeze force is a one-dimensional scalar computed from the full contact reactions:

$$F_{\text{squeeze}} = \min\left(\|\mathbf{f}_L^{S_L}\|,\ \|\mathbf{f}_R^{S_R}\|\right) \in \mathbb{R}_{\geq 0}. \tag{2}$$

We assume the object is at the threshold of relative sliding while remaining stationary. Because the pressure directions at the two fingertips are not necessarily collinear, we define the force required to keep the object stationary, F_{static}, in terms of the grip force F_{grip}, the friction coefficient μ, and the squeeze proxy as

$$F_{\text{static}} = F_{\text{grip}}\mu = 2F_{\text{squeeze}}\mu \approx \|\mathbf{f}_L^T + \mathbf{f}_R^T\|\mu. \tag{3}$$

The TCP frame is denoted by T. Lowercase $\mathbf{f}_L^T, \mathbf{f}_R^T$ are the left and right sensor reactions expressed in this common frame, and capital $\mathbf{F}^T$ denotes the single resultant force exerted on the object. Rotating each sensor reaction into T and applying action–reaction gives

$$\begin{aligned} \mathbf{f}_L^T &= \mathbf{R}_{S_L}^T \mathbf{f}_L^{S_L}, \qquad \mathbf{f}_R^T = \mathbf{R}_{S_R}^T \mathbf{f}_R^{S_R}, \\ \mathbf{F}^T &= -(\mathbf{f}_L^T + \mathbf{f}_R^T). \end{aligned} \tag{4}$$

The sensor-local normal axes map to opposite directions along TCP y. The scalar aperture p is the distance between the two tactile contact planes along y. Approximating the contacts by the pad centers, at $(0, -p/2, 0)^\top$ and $(0, p/2, 0)^\top$ relative to the TCP, gives the moment directly from the two forces:

$$\boldsymbol{\tau}^T \approx \frac{p}{2}\left(f_{L,z}^T - f_{R,z}^T,\ 0,\ f_{R,x}^T - f_{L,x}^T\right)^\top. \tag{5}$$

We assume the grip force is sufficient to maintain stable contact without relative slip between the object and tactile pads. Under this assumption, torque about the grasp axis y still depends on the object and its contact conditions. To keep the calculation centered on the sensors and known gripper geometry, we leave this object-dependent component outside the modeled wrench and its feedback correction. The zero y-moment entry therefore marks an omitted component, not a physically zero torque. Figure 3 illustrates the decomposition across gentle grasping, blackboard wiping, card swiping, and knob turning.

Wrench-Aware Control The policy predicts an end-effector pose $\mathbf{P}^{\text{pred}} \in SE(3)$ and target sensor reactions in the two local frames. The controller rotates each target and measurement into the TCP frame using rotation matrices before subtraction:

$$\begin{aligned} \mathbf{e}_L^T &= \mathbf{R}_{S_L}^T(\mathbf{f}_L^{S_L,\text{target}} - \mathbf{f}_L^{S_L}), \\ \mathbf{e}_R^T &= \mathbf{R}_{S_R}^T(\mathbf{f}_R^{S_R,\text{target}} - \mathbf{f}_R^{S_R}). \end{aligned} \tag{6}$$

The sum and difference of the two fingertip errors give the force-driven pose corrections directly:

$$\begin{aligned} \Delta\mathbf{p} &= -\mathbf{K}_p \mathbf{R}_T^B(\mathbf{e}_L^T + \mathbf{e}_R^T), \\ \Delta\boldsymbol{\phi} &= \mathbf{K}_\phi \mathbf{R}_T^B \begin{bmatrix} \frac{p}{2}(e_{L,z}^T - e_{R,z}^T) \\ 0 \\ -\frac{p}{2}(e_{L,x}^T - e_{R,x}^T) \end{bmatrix}, \\ \mathbf{p}^{\text{cmd}} &= \mathbf{p}^{\text{pred}} + \Delta\mathbf{p}, \qquad \mathbf{R}^{\text{cmd}} = \exp\big([\Delta\boldsymbol{\phi}]_\times\big)\mathbf{R}^{\text{pred}}. \end{aligned} \tag{7}$$

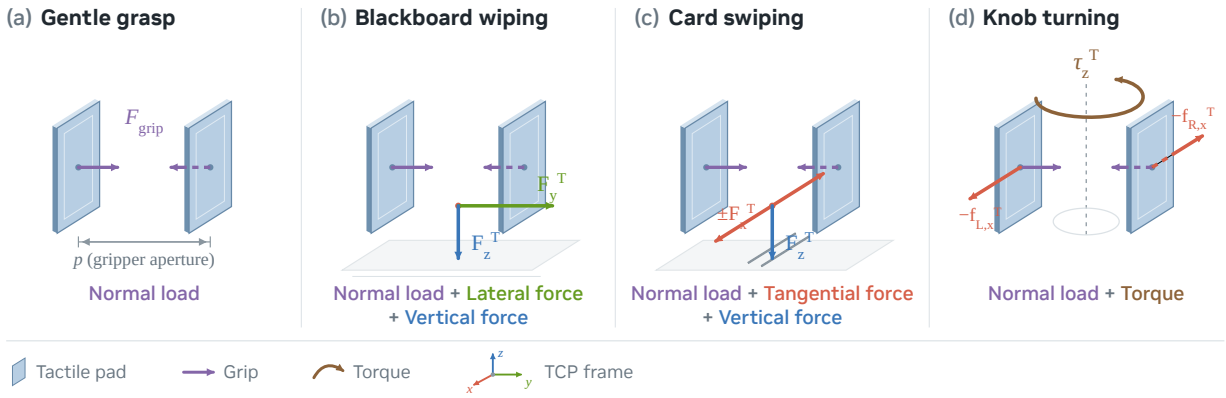


Figure 3: **Fingertip-force decomposition across tactile skills.** Purple arrows denote the internal grip channel and colored arrows the TCP forces and torque; every arrow is drawn as a load on the object, dashed where a pad hides it. (a) Gentle grasping, (b) blackboard wiping, (c) card swiping, and (d) knob turning.

Here $\mathbf{R}_T^B$ rotates from TCP to base coordinates while moments remain referenced to the TCP. The gains $\mathbf{K}_p$ and $\mathbf{K}_\phi$ have units $\mathrm{m/N}$ and $\mathrm{rad/(N \cdot m)}$, respectively, and $[\cdot]_\times$ denotes the cross-product matrix. The policy first predicts a position, then corrects it based on force feedback. The rotation vector $\Delta\phi$ updates orientation through the exponential map. Differential inverse kinematics converts the corrected pose into joint velocity commands, while the independent grip loop adjusts aperture.

Grip Force Control To overcome the limitation of position control in precise force regulation, we implement a grip force feedback loop. The predicted sensor-frame reactions $\mathbf{f}_i^{S_i,\text{target}}$ provide the contact force magnitudes directly, giving the squeeze proxy and then $F_{\text{grip}}^{\text{pred}}$ through Eqs. (2) and (3); the grip-loop target is

$$F_{\text{grip}}^{\text{target}} = (1 + k_{\text{ff}})\, F_{\text{grip}}^{\text{pred}}, \quad k_{\text{ff}} > 0, \tag{8}$$

where k_{ff} is a feedforward gain. The measured grip force $\tilde{F}_{\text{grip}}$ is exponentially smoothed. Let p denote the gripper width. We apply a correction Δp only when $|\Delta p| \geq \text{dz}$, where $\text{dz} > 0$ is a deadzone threshold accounting for gripper imprecision:

$$\Delta p = k_p^{\text{adm}} \cdot (F_{\text{grip}}^{\text{target}} - \tilde{F}_{\text{grip}}), \tag{9}$$

with k_p^{adm} an admittance gain. The final gripper command is

$$p^{\text{cmd}} = \text{clip}\big(p^{\text{pred}} + \Delta p,\ 0,\ p_{\max}\big), \tag{10}$$

where p^{pred} is the policy-predicted gripper width. This allows the policy to specify a desired grip force while the controller achieves accurate tracking via feedforward and force feedback. In summary, the two fingertip force triples together with the aperture provide the signals used by the low-level controller: a grip-force proxy from the contact force magnitudes, and estimated wrench-related pose corrections from the sum and the half-aperture-scaled difference, while task-level coordination is delegated to the policy.

3.2.3. Tactile Policy

Tacinf-VTLA is trained on Tacinf-Tabero data and consumes the tactile representations and force-aware instructions produced by the infrastructure. To integrate tactile signals into the VLA foundation model, we evaluate three tactile input adapters that map different tactile modalities into conditional tokens. The high-level policy then jointly predicts future gripper poses and six-dimensional fingertip-force setpoints based on vision, language, and touch. These commands are tracked by the fixed admittance-based low-level controller described above (Fig. 4), which provides physical compliance; the high-level policy learns to reason about appropriate

interaction forces given the task context. Building on the OpenPi infrastructure Black et al. (2025b) and leveraging flow matching, our approach enables continuous prediction of both pose and the six fingertip force components. The three adapters share the same temporal window, preprocessing, backbone, and training protocol; only the tactile input interface is varied in the ablation. Below, we detail the adapters and loss function, and compare alternative tactile injection strategies inspired by prior work.

Marker-Field Adapter. It processes tactile marker displacement fields relative to the initial rest state of the sensor. The input consists of $H+1$ frames: the first frame captures the undeformed marker layout, and the subsequent H frames record 2D marker positions in the sensor's local plane, resulting in an array $M \in \mathbb{R}^{(H+1)\times N\times 2}$. Here, N denotes the number of markers, and the 2D element denotes the marker's planar coordinates. A lightweight Temporal Convolutional Network (TCN) encodes this spatiotemporal sequence into tokens for integration into the transformer backbone. Detailed model configurations will be released with the code; Appendix A summarizes the evaluated variants.

Tactile-Image Adapter. Inspired by visual–tactile policy conditioning Xue et al. (2025); Zhang et al. (2026), we implement an image adapter for our backbone. Specifically, we construct a single composite tactile image by arranging the H historical frames from the left finger and the H frames from the right finger into a unified spatial layout. The value of H is determined by the shared multi-view image resolution and the prescribed tactile-image resolution; when $H=1$, the composite contains only the current contact state. This image, encoding the full tactile history of both fingertips, is processed by the same visual encoder used for RGB-D observations. Its features then interact with visual features via cross-attention in the transformer, enabling joint reasoning over contact and scene geometry. Implementation details and parameter settings will be released with the code.

Force Adapter. We use the 3D sensor-frame reaction vectors $\mathbf{f}_L^{S_L}$ and $\mathbf{f}_R^{S_R}$ as raw inputs, resulting in a 6-dimensional force signal. Although these reactions can be transformed and combined with the known fingertip geometry to compose the applied wrench at the TCP, we find it more effective to directly feed their concatenation into a multilayer perceptron (MLP) to obtain a compact latent representation of contact dynamics. This representation is injected into the model alongside other modalities to enhance physical awareness. Specific network configurations and training parameters will be released with the code.

Force Supervision. Flow matching Black et al. (2025b) supports continuous-valued outputs, including force components, motivating our weighted loss design. The action vector a includes both pose commands and six sensor-frame force components, $[\,\mathbf{f}_L^{S_L}\,;\,\mathbf{f}_R^{S_R}\,]$. For a batch b and time $t\sim\mathcal{U}(0,1)$, we define the interpolated input $x_t=(1-t)\epsilon+ta$ and target velocity $u_t=a-\epsilon$, where $\epsilon\sim\mathcal{N}(0,I)$. The prediction error $e=v_t-u_t$ is split into action and force components, $[e^{(\text{act})}, e^{(\text{force})}]$. We normalize the six force dimensions to a common scale and upweight them by a factor $\lambda_{\text{force}}>0$, while keeping other dimensions at unit weight. The final loss is:

$$\mathcal{L}=\frac{1}{D_{\text{act}}}\sum_{d=1}^{D_{\text{act}}} w_d(e_d)^2, \tag{11}$$

where $w_d=\lambda_{\text{force}}$ for dimensions corresponding to the predicted fingertip forces, and $w_d=1$ otherwise, with $\lambda_{\text{force}}>0$. Although the six components share the same unit (N), their magnitudes differ across directions, so the normalization is necessary for stable six-dimensional supervision.

3.3. Tacinf-Tacumi

We develop Tacinf-Tacumi, a handheld multimodal interface for collecting demonstrations of contact-rich manipulation. Following the robot-free teaching paradigm of UMI Chi et al. (2024), the system combines tactile sensing with real-time active force feedback, allowing operators to perceive and regulate grasping load while directly manipulating objects. It synchronously captures end-effector motion, gripper aperture, and contact observations to provide multimodal demonstrations for tasks requiring fine force regulation.

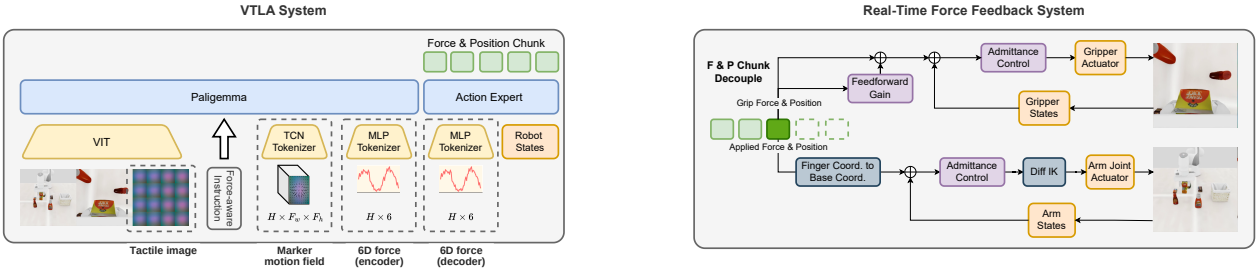


Figure 4: **Tacinf-VTLA system overview.** Tactile inputs are encoded by input adapters and fused with vision and language. The policy predicts force–position commands, which the two-loop closed-loop controller tracks to achieve compliant interaction.

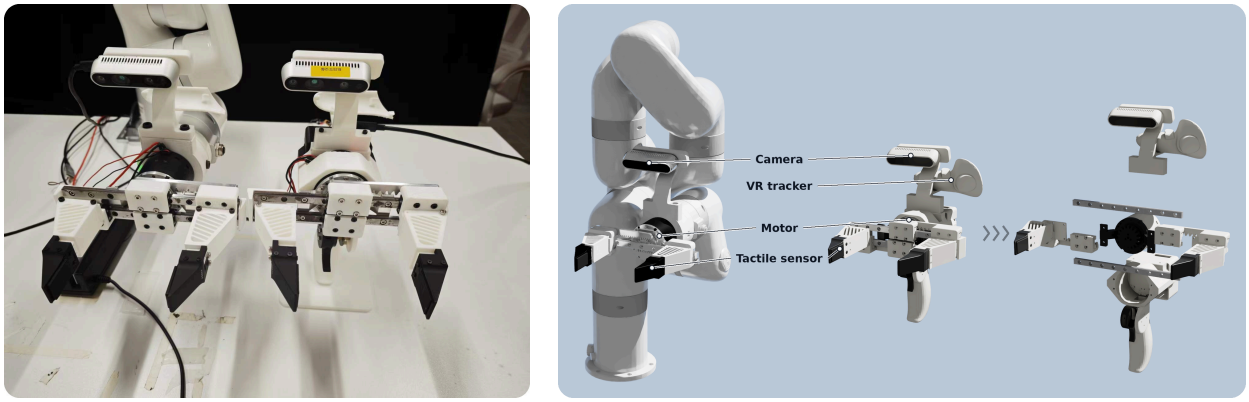


Figure 5: **Tacinf–Tacumi hardware.** Left: the real handheld collection device, with its QDD drive, wrist RGB camera, and tactile fingertips. Right: structural render, at one common scale, of the arm-mounted gripper, the handheld device, and the handheld device taken apart, with the parts the two sides share labelled (camera, tactile sensor, motor) and the VR tracker that the handheld device alone carries for UMI pose tracking.

As shown in Figure 5, Tacinf-Tacumi comprises a motorized trigger, a parallel gripper, and multimodal sensing components. The trigger operates under field-oriented control (FOC) and serves as both the opening–closing command interface and the active feedback actuator. The parallel gripper is driven by a DAMIAO motor that provides position and load feedback, while fingertip-mounted Xense vision-based tactile sensors provide tactile images and contact-force estimates. A VR controller supplies six-degree-of-freedom pose tracking, and a camera records visual observations. The acquired signals and task instructions are organized under a common data schema, allowing handheld demonstrations to be combined with robot teleoperation and simulation trajectories for the mixed-data training evaluated in Section 4.8.

3.3.1. Fine Force Manipulation

The handheld interface establishes a bidirectional interaction loop through position commands and torque feedback, following the position–force organization of bilateral interfaces Lawrence (1993). In the command channel, trigger rotation is mapped to a target gripper aperture and converted into a motion command for the DAMIAO motor. The actual aperture is obtained from motor position feedback through the transmission relationship. The target and actual apertures represent operator intent and gripper execution state, respectively.

In the feedback channel, the torque estimate reported by the gripper motor is mapped to a feedback command

for the trigger:

$$\tau_{\text{trigger}}^{\text{cmd}} = k_{\text{fb}} \hat{\tau}_{\text{gripper}}, \tag{12}$$

where $\hat{\tau}_{\text{gripper}}$ denotes the estimated gripper motor torque and k_{fb} is a signed feedback gain. Its direction follows the coordinate conventions of the two motors so that the grasping load produces a reaction torque opposing further trigger closure. The trigger motor executes this command through FOC Bhardwaj (2013), rendering changes in gripper load as resistance perceptible to the operator and supporting continuous force regulation through real-time interaction feedback.

Motor torque feedback and tactile force estimation serve complementary roles. Motor torque includes contributions from transmission friction and actuator dynamics Le Tien et al. (2008); it is therefore used to convey load variations rather than as a direct measurement of fingertip contact force. The Xense sensors provide contact-force estimates using the manufacturer's calibration, supporting contact observations, real-robot force statistics, and the grip-force feedback controller in Section 3.2. This design combines active feedback for the operator with contact sensing for policy learning and robot control while preserving the distinct physical meanings of the two signals.

3.3.2. Data Collection

The system separately acquires the gripper's six-degree-of-freedom rigid-body pose and single-degree-of-freedom aperture to represent end-effector motion and grasp state. The tracking design considers trajectory reliability, system integration complexity, and deployment overhead across collection environments.

Existing handheld collection systems employ different pose-acquisition strategies. UMI estimates trajectories from wrist-camera observations and inertial measurements, but its original paper acknowledges a dependence on sufficient environmental texture, with limitations in texture-deficient settings such as rooms with plain white walls Chi et al. (2024). TacUMI integrates a VIVE Tracker into the handheld device, providing an example of acquiring manipulation trajectories with a dedicated tracking device Cheng et al. (2026a). For Lighthouse-based implementations such as VIVE Tracker 3.0, external base-station installation and spatial configuration introduce additional preparation when moving between collection sites HTC VIVE (n.d.). ViTaMIn-B instead adopts Meta Quest controllers and reports a higher proportion of valid demonstrations than the SLAM-based approach used in UMI on its Weight Placement task, providing empirical support for consumer VR tracking in handheld collection Li et al. (2025a).

Building on these precedents, we adopt the commercial headset-and-controller tracking system of Meta Quest 3, separating end-effector pose acquisition from wrist-camera trajectory reconstruction. This solution provides real-time poses without external tracking base stations, reducing site configuration requirements and facilitating portable demonstration collection across everyday environments. The controller is rigidly attached to the handheld device, and its tracked pose is transformed into the gripper tool center point (TCP) pose through a fixed extrinsic transformation, consistent with the controller-to-end-effector mapping used in prior handheld interfaces Li et al. (2025a):

$$^{W}\mathbf{T}_{\text{TCP}}(t) = {}^{W}\mathbf{T}_{C}(t)\, {}^{C}\mathbf{T}_{\text{TCP}},$$

where W and C denote the tracking reference frame and controller frame, respectively, and ${}^{C}\mathbf{T}_{\text{TCP}}$ is the fixed controller-to-TCP transformation. The controller-to-gripper-center transformation is estimated from 20 paired robot and controller poses using hand–eye calibration, as detailed in Appendix E.1.

Gripper aperture is computed from motor position feedback through the transmission relationship, allowing actuation and state acquisition to share the same hardware. End-effector pose, gripper aperture, control commands, load feedback, and visual and tactile observations are aligned on a common timeline to form multimodal trajectories that preserve the correspondence between actions and contact information. The native acquisition rates and common host timestamp convention are described in Appendix E.2.

3.3.3. Low Cost and Actuator Consistency

Prior interfaces address collection-to-deployment differences through hardware design, calibration, or adaptation Wi et al. (2026); Xu et al. (2025a); Yu et al. (2025a); Zhang et al. (2025a). Tacinf-Tacumi uses the same low-cost gripper motor for handheld collection and robot execution, reusing the gripper actuation design and its command and state interfaces. During deployment, the robot operates the corresponding motorized gripper, while the trigger serves only as the human interface during demonstration collection. This design avoids selecting a different, dedicated gripper motor for deployment, reducing hardware selection, control-interface adaptation, and system integration effort.

Reusing the actuation design helps reduce control and calibration differences introduced by actuator replacement. Its cost advantage derives from the low-cost motor selection and design reuse across collection and execution. This does not imply identical contact responses: transmission geometry, fingertip geometry, sensor mounting, and control parameters still require matching or calibration. Within this scope, the design reduces additional hardware and engineering requirements for deployment while providing a basis for consistency between the demonstration data and the robot execution interface.

3.4. Tacinf-Taco

Tacinf-Taco (Tactile policy Co-training) combines teleoperation, simulation, and Tacumi (UMI) demonstrations for VTLA post-training. Simulation data are generated by replaying teleoperation trajectories in task-aligned Tacinf-Tabero scenes. Tacumi demonstrations provide a parallel real-data source with varied object positions, orientations, and approach motions. All three sources are mixed to train one policy. Collection effort decreases from teleoperation to Tacumi to simulation. Figure 6 summarizes these sources and their relation.

The framework consists of three dependent parts:

1. **Task-aligned scene reconstruction:** transfer workspace appearance and interactive-object geometry into a common task frame, providing the assets for controlled visual variation.
2. **Manipulator-system Real2Sim:** align the robot, cameras, tactile sensing, and action interfaces so that replay produces synchronized observations and force targets compatible with real data.
3. **Augmentation and post-training:** combine simulation-based visual and force-level variation with UMI-based spatial variation, then return the resulting VTLA policy to the shared physical interface.

The aligned simulator also supports policy evaluation. The simulation-only adaptation protocol and paired Sim/Sim2Real evaluation are specified in Sections 4.5 and 4.6; they are separate from the mixed-data post-training design described here.

The reusable unit is the interface contract rather than a particular asset or backbone. A new source dataset is mapped to the task/state/success schema, a new tactile sensor supplies a calibrated observation model, and a new pretrained VLA supplies its own input adapter and action head while retaining the common timestep, normalization, force-supervision, and evaluation interfaces. This makes the framework extensible by design.

Following real-to-sim evaluation approaches that align policy-relevant observations and control Jain et al. (2026); Li et al. (2025b), and tactile sim-to-real studies such as AnyRotate and SimShear Freud et al. (2025); Yang et al. (2025), Taco uses two practical criteria. First, **visual–physical separation**: Gaussian or neural appearance models are used only for rendering and never for contact; explicit collision geometry handles contact, support, and shadow reception. Second, **policy-relevant consistency rather than absolute-parameter equality**: parameters are adjusted to match the data structure, distribution, and visual effect that matter to the policy, because real physical parameters often require time-consuming and extensive calibration and measurement, and some are unavailable under commercial restrictions. In practice, we check contact onset, contact-patch persistence, force scale, marker response, and action replay. These checks distinguish our protocol

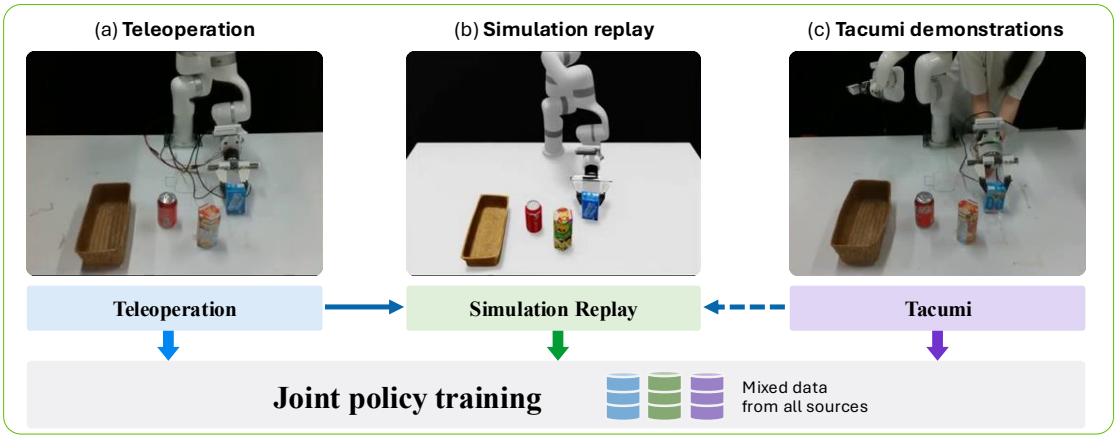


Figure 6: **Tacinf–Taco data sources and co-training.** (a) Real external-camera view of the teleoperated task. (b) Simulation replay of the same trajectory in the configured Tacinf-Taco scene, with Vitasoy, Coca-Cola, cookie, and basket. (c) Tacumi demonstration: the UMI camera view at the moment of grasping the Vitasoy box. Simulation replays teleoperation trajectories; Tacumi demonstrations are collected in parallel. Training uses all three sources.

from a claim of pixel-level digital-twin fidelity.

3.4.1. Task-Relevant Scene Reconstruction

This module transfers the real workspace and interactive objects into Isaac Sim, enabling rendering and interaction under the same camera extrinsics and operation region. The workspace is represented visually by 3D Gaussians, while objects enter the physical layer as independent textured meshes. Both are aligned to the task coordinate system and obey the visual–physical separation principle.

Task-Aligned Workspace Reconstruction. We represent workspace appearance with 3D Gaussian Splatting (3DGS) Kerbl et al. (2023) and convert it into an Isaac Sim-loadable NuRec Volume. The task-side robot, white collision table, cameras, and operation region remain unchanged; the Gaussian asset is aligned to the task frame using a single registration matrix $\mathbf{T}_{\text{reg}}$. We use the Odin1 spatial camera from Liuxing Technology for reconstruction data capture, because in addition to a directly usable video stream, its built-in LiDAR odometry provides a more accurate reference for camera extrinsics. A spatial camera continuously records trajectories around the operation region that cover the reachable space and fields of view of the third-person and wrist cameras, while logging calibrated images, poses, and COLMAP intrinsics/extrinsics Schonberger and Frahm (2016). The reconstruction is exported as a standard 3DGS point set in a Z-up, meter-based convention. The capture focuses on the frustum of the third-person camera actually used by the policy, together with the frusta swept by the wrist camera as it moves across the full workspace of the manipulator arm, rather than uniformly covering the entire room.

3DGS Post-Processing. During scanning, the robot arm is present in the scene, so retaining its Gaussians would create double geometry with the simulated URDF robot. Under a tabletop-plane constraint, we perform RANSAC plane estimation Fischler and Bolles (1981), candidate-point selection with tabletop protection, connected-component clustering and morphological filtering, and multiple cleanup passes based on bounding-box neighborhoods, nearest-neighbor growth, and base-column constraints, while protecting fine structures such as cables wrapped around the arm. For rendering compatibility, we remove far-field outliers, contract higher-order spherical harmonics to DC-only, and ensure that the declared SH order matches the specular tensor dimensions, zero-padding missing dimensions when necessary. Finally, the scene USD uses an outer

transform to cancel the task's fixed external frame and retains only $\mathbf{T}_{\text{reg}}$ in the inner transform relative to the robot base.

Object Reconstruction. Interactive objects enter the task as independent USD assets rather than being extracted from workspace Gaussians. For each real object, we capture one image and use Seed3D 2.0 Gu et al. (2026) to generate a textured 3D model exported as USD. The explicit mesh follows Isaac's rasterization path; when Gaussian appearance and an object mesh overlap, the object asset owns the interaction while the Gaussian layer supplies only unmodeled background. The exported USD has no collision geometry. We therefore add rigid-body properties through a USD overlay and generate collision meshes with convex decomposition without modifying the generated asset itself. The visual mesh is used only for rendering and the collision mesh only for contact; object physics is supplied by the overlay. Because Seed3D scale and initial pose can differ from the real object, we import the asset into Isaac Sim, replay the recorded real demonstrations, and iteratively adjust object scale and initial position along each axis until grasp and placement trajectories align. The calibration objective is to improve trajectory and success-rate consistency between simulated replay and the real robot, rather than to minimize appearance error alone; during this calibration stage, task demonstrations and robot trajectories remain unchanged. Before evaluation, we check the reconstructed scene with task initialization and camera observations.

3.4.2. Embodiment Real2Sim

Scene alignment alone is insufficient for policy transfer: the simulated embodiment must expose the same action and observation semantics as the real system. To ensure the generality of our framework, we do not reconstruct the robot arm's complete Gaussian appearance; instead, we reproduce only its dynamics and kinematics system. In addition, we explicitly simulate the shared Tacumi gripper, including its gripper control and tactile sensing.

Manipulator System. We build the XArm dynamics in Isaac Lab and tune its PD parameters to match the real robot's control behavior. Hand–eye calibration provides the third-person camera pose relative to the robot base. Because the Tacumi gripper uses the shared gripper/sensor assembly, the wrist-camera relationship is known from its mechanical construction rather than estimated from an external sensor model. Finally, we tune the shared tactile gripper's passive-joint mode, PD parameters, and the force–position hybrid controller introduced above.

Tactile Observation Modeling. Our tactile Real2Sim path combines the Taxim optical model Si and Yuan (2022) and the FOTS marker-motion model Zhao et al. (2024) in a shared Isaac Lab sensor interface. The simulator first derives a contact representation from the depth image beneath the pad. Taxim maps local surface geometry to a tactile RGB observation, while FOTS combines contact deformation with contact-centroid changes and relative sensor–object yaw to produce a marker displacement field in sensor image coordinates. The two models are therefore complementary forward models: they reproduce the signals exposed to the policy without requiring an exact replica of the commercial sensor's internal optics or elastomer mechanics. Following the calibration methods of Taxim and FOTS, we calibrate the above models using a small set of real tactile observations and, through parameter settings in Isaac Sim, align the simulated and real data streams in resolution, channel semantics, preprocessing, and temporal-update conventions. We jointly adjust the observation mapping and the contact geometry. In particular, the normal thickness of the tactile-pad collision volume is treated as an asset-level variable and tuned to reduce discrete-time interpenetration while preserving a stable contact patch and consistent activation of both tactile models. This calibration targets policy-relevant contact responses rather than visual mesh agreement alone. Appendix D details the depth-to-indentation mapping, optical and marker interfaces, and calibration scope; the complete simulator configuration is released with the calibrated asset and pinned by the run manifest. Due to commercial restrictions, we cannot obtain the complete optical and mechanical specifications of Xense. We therefore only tune the cameras' rendered view and operating depth range at the observation level, and do not claim strict structural extrinsic registration.

3.4.3. Data Augmentation and Joint Post-Training

Taco combines three data sources for co-training: teleoperation data, Tacumi data, and simulation data. The goal is to improve the policy's generalization on real-robot tasks while preserving consistent language, contact, and action semantics across sources.

Real seeds and simulated visual augmentation. Real-robot trajectories supply the original task instructions, motions, and available observations, and anchor the scene and sensor calibration. Object positions remain relatively fixed, which yields better simulation alignment; the missing generalization is supplied by UMI data. Starting from the aligned assets, the simulation branch varies object appearance or instances and background visual elements. Building on Tacinf-Tabero, we replay the task under these variations and controlled force regimes, producing synchronized visual observations, tactile images, marker fields, contact forces, and action targets. Background appearance changes leave the collision model unchanged; replacing an interactive object's geometry requires a corresponding collision asset and a new contact rollout. Tactile signals and force labels therefore come from replay in the varied scene, rather than being copied onto edited images. After replay, a contact-signal check and a task-completion check filter out data that produced no contact or failed the task.

Spatial augmentation with UMI. Tacumi extends spatial coverage with human demonstrations. Objects are placed at different positions and orientations and approached along different paths. The shared gripper records the resulting changes in contact and motion together, adding physical spatial variation to the simulated visual and force-level variations. This branch collects new demonstrations of the task variants; changing an object's pose is not treated as an image-only transformation of an existing action sequence. Tracked UMI poses and sensor observations are expressed through the common task and observation–action interfaces before mixing.

Joint VTLA post-training. The real, simulated, and UMI trajectories are synchronized, normalized, and sampled into mixed batches to jointly post-train a single pretrained Tacinf-VTLA policy with the action and fingertip-force supervision in Section 3.2. The source sampling ratio and task/force-level coverage are explicit training settings. The post-trained policy is deployed with the same tactile gripper and hybrid controller.

Data synergy. Among the three data sources, simulation data are the cheapest, UMI data come next, and teleoperation data are the most expensive. Simulation data provide visual-element and object generalization, UMI data provide grasp-approach and object-pose variation, and teleoperation data provide precise examples. The co-training study in Section 4.8 evaluates this combined-data loop: the three-source mixture substantially outperforms teleoperation-only training on out-of-distribution (OOD) tasks, and the characteristics of the simulation and UMI data are reflected in the policy's capabilities—without UMI data, position generalization is hard to achieve; without simulation data, object generalization is hard to achieve; without teleoperation data, the accuracy of the UMI and simulation data decreases; and UMI-only data cannot achieve good results. We summarize this as a data synergy effect.

4. Evaluation

Sections 4.1–4.3 evaluate nine tactile-input variants of the foundation model on Tabero; Sections 4.4–4.6 evaluate eight models retrained with Taco data; Section 4.7 evaluates one model on separate controller data; and Section 4.8 evaluates six models retrained with different data mixtures. The hardware consists of one XArm7 arm equipped with an Xense-based tactile gripper and its UMI counterpart for data collection, together with a Meta Quest 3 for UMI pose tracking. Compute devices include four H20 and two A6000 GPUs; a single RTX 4090 workstation is the minimum compute requirement for reproducing the complete system.

Evaluation protocol. Counting the two force levels as distinct task requirements, Tabero contains 18 base tasks and Taco contains six base tasks, six position-randomized tasks, and six tasks with both position and object randomization; three additional manipulation tasks are used to validate the controller. The task categories are language-conditioned force following and special-object manipulation. Each base Tabero and Taco task has 30

demonstrations; data quantities for mixed-data and standalone task experiments are specified separately. In simulation, each model is evaluated for 50 trials per task. On the real robot, each model is evaluated for 20 trials on force-following tasks and 10 trials on special tasks, using the criteria in Appendix B.2. Sections 4.1–4.3 use the per-finger minimum in Eq. (2) as grip force. Sections 4.4–4.8 use the sum of left and right fingertip force magnitudes as an approximation because individual fingertip readings are inaccurate during contact; see Appendix C.4. A trial is terminated when it exceeds the maximum episode length. Firm and gentle conditions of the same task share identical initial object and robot states. The real-robot policy runs at 20 Hz and executes actions asynchronously through temporal ensembling, with an inference latency of roughly 150–200 ms; the tactile closed-loop controller runs at the same 20 Hz rate.

4.1. Data Reuse and Annotation

To validate the effectiveness of the Tacinf-Tabero data migration and reuse method, we select LIBERO, a representative MuJoCo-based open-source dataset, as the data source. We validate the fidelity of data migration from MuJoCo to Isaac Lab by evaluating both task success rates and distributional consistency. Specifically, we select four subtasks from the LIBERO benchmark suite and compare the success rates of the original MuJoCo-based dataset with those of our replayed version in Isaac Lab. Fig. 2 illustrates the range of task scenarios our simulation platform can provide. In subsequent experiments, we only used a subset of these configurations and did not randomize the visual modality.

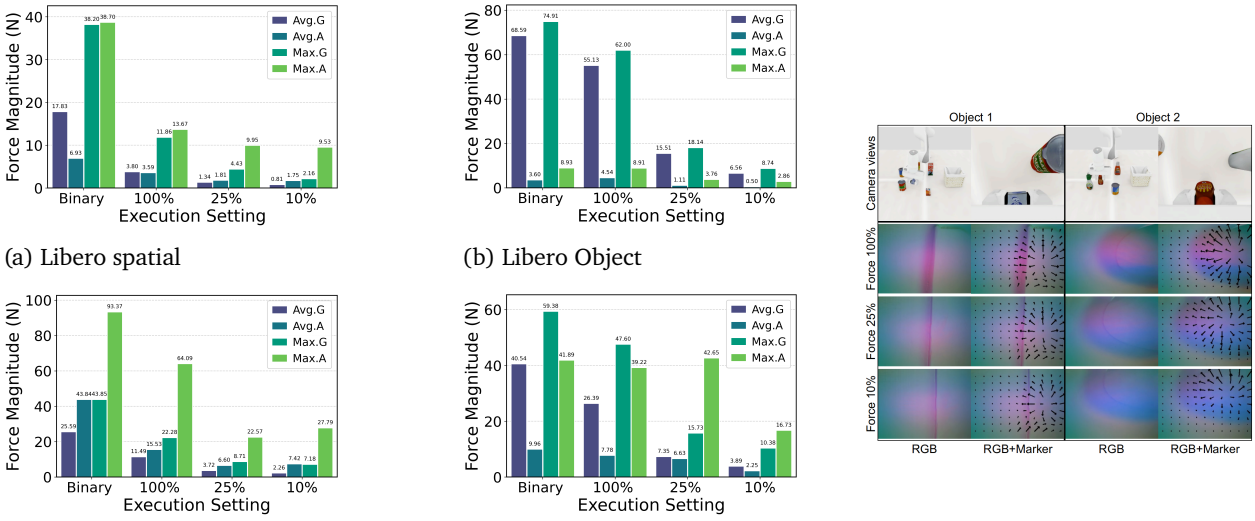
When using the same robot kinematics and control policy as in the original dataset, our baseline configuration yields a success rate distribution that closely matches that reported in OpenVLA Kim et al. (2025b). However, replacing the standard parallel-jaw end effector with a tactile-sensor-integrated gripper on the same Franka arm introduces mechanical differences that lead to a measurable drop in success rates. This degradation is especially pronounced in tasks requiring delicate manipulation, where lower grip forces strongly correlate with reduced success. The results shown in Table 1 highlight the sensitivity of force-aware tasks to end-effector design and force regulation.

Table 1: Cross-platform data validation: task success rates across four LIBERO subtasks. We compare the original MuJoCo dataset, our replay in Isaac Lab with identical robot configuration (denoted Isaac), and our modified setup with a tactile-equipped Franka gripper at 100%, 25%, and 10% force scaling (denoted T-100, T-25, and T-10).

Task	MuJoCo	Isaac	T-100	T-25	T-10
Spatial	0.86	0.83	0.42	0.24	0.07
Object	0.91	0.77	0.84	0.87	0.73
Goal	0.76	0.78	0.55	0.44	0.30
LIBERO-10	0.86	0.66	0.60	0.47	0.32
Average	0.85	0.76	0.60	0.50	0.36

4.2. From Discrete to Continuous Force-Position Control

We verify whether our data collection framework effectively expands the distribution of interaction forces. We compare a baseline using binary gripper control, which logs only discrete open/close commands and regulates no force, against our approach, which explicitly sets different force parameters during execution; results are shown in Fig. 7. Specifically, we define 100% grip force as the “firm” condition, modifying the original instruction with the adverbs “firmly” and “tightly”, and 25% and 10% as the “gentle” conditions, annotated with “gently” and “softly”. This is an ordered scale rather than a binary distinction: the same adverb set maps monotonically to increasing grip force with the commanded level, and the out-of-domain adverbs



(a) Libero spatial

(b) Libero Object

(c) Libero Goal

(d) Libero 10

Figure 7: **Force distribution and tactile observations at different force magnitudes. Left:** force distributions under various control modes across different task suites. “Binary” represents the binarized control commands applied by a non-tactile gripper; “100%”, “25%”, and “10%” indicate the force distributions of a tactile gripper under different force settings, where “100%” retains the default gripper PD gains of the original continuous-control setup. The two are not the same setting: “Binary” logs only discrete open/close commands and regulates no force target, which is why its peak applied force exceeds that of the 100% level. **Right:** tactile images and corresponding camera images at different force magnitudes. The left two columns represent the first category of objects and the right two columns the second; RGB denotes the simulated tactile RGB image, and RGB+Marker overlays the simulated marker motion field on the RGB image. In the camera views, the left view is from the third-person camera and the right view is from the wrist-mounted camera.

reserved for testing (Appendix C.2) fall between the two extremes rather than collapsing onto either one, so the language condition specifies a position on a continuous force axis. “Binary” refers only to the baseline command interface, not to a third language condition. In our experiments, we set $K_p \in \{2000, 500, 200\}$ N/m and $K_d \in \{100, 25, 10\}$ N·s/m to correspond to 100%, 25%, and 10% force levels, respectively, where the 100% setting follows the default parameters for the Franka robot in Isaac Sim. For the same task, we present tactile images, marker fields, and contact force measurements at comparable contact stages under these distinct force settings in Fig. 7, demonstrating clear multimodal variations aligned with the intended interaction intensity.

Furthermore, we observe that under the extreme 10% force setting, the retention rate of most tasks is relatively low. Appendix C gives the formal definition of the retention rate. Therefore, in subsequent ablation tests, we constructed a Tabero subset to analyze the policy’s performance under extreme conditions. This subset includes 9 tasks from the Object dataset, each executed under two force conditions specified by linguistic adverbs. The composition of the dataset is presented in Appendix B.1. We collect three datasets over the task group, differing only in force magnitude and gripper actuation strategy. **Dataset A** uses continuous control with “gentle” forces at 25% of the corresponding “firm” level. **Dataset B** also employs continuous control but reduces the force to 10%, representing an extreme low-force regime where slippage is likely. **Dataset C** uses the same 10% force level but switches to binary open/close commands, with only discrete gripper states logged.

During evaluation, the arm and gripper operate with impedance parameters corresponding to 100% force scaling. The Tacinf-VTLA policy predicts target contact forces conditioned on language and visual inputs; these targets are tracked by a closed-loop admittance controller that modulates impedance during execution. We adapt a base VLA model using LoRA to incorporate tactile marker fields (Dataset A and B), while a vision–

language-only variant is trained on Dataset C for ablation, where the binary gripper commands carry no continuous force target to predict. As shown in Table 2, that variant does not modulate force at all (AG 35.8 N under firm against 34.6 N under gentle), consistent with the ablation below: force modulation appears only when the policy is given a continuous force target to predict. Furthermore, the sharp drop in success rate from 25% to 10% force scaling reflects the increased difficulty of maintaining stable contact under ultra-gentle constraints, underscoring that Dataset B represents a substantially more challenging regime for gentleness-aware policies. Appendix C.2 reports the corresponding language-to-force trend on this dataset, including the out-of-domain adverbs.

Table 2: Training results on the three datasets. Each cell gives the firm / gentle value, where the gentle condition is 25% for Dataset A, 10% for Dataset B, and 10% with binary gripper commands for Dataset C.

Metric	A 100% / 25%	B 100% / 10%	C 100% / 10%
SR	0.87 / 0.79	0.86 / 0.52	0.86 / 0.87
AG	31.3 / **8.5**	32.4 / **3.7**	35.8 / 34.6
MG	61.3 / 19.9	62.7 / 11.4	65.7 / 65.0

4.3. Ablation and Comparison of VTLA

To compare and conduct ablation experiments on different tactile injection methods and force control strategies, we evaluate tasks from **Dataset B**. Among them, Force E+FS, Img, and Force D+FS are adaptations of the design principles of Huang et al. (2025c), Zhang et al. (2026), and Zhang et al. (2025b) respectively, re-implemented within our shared backbone and training protocol. They are not reproductions of the original systems: the backbone, data, and training protocol are ours, so the numbers reported here are not comparable with the values those papers report. What the comparison tests is how these published design choices behave under one shared backbone, dataset, and protocol, and the results below should be read in that scope. Except where the ablation architecture requires a different tactile interface, all models were fine-tuned via LoRA Hu et al. (2022) with an identical set of hyperparameters; detailed training and model configurations will be released with the code.

Fig. 8 shows that the baseline without tactile input and without force supervision fails completely. The ablation then separates the two factors, and the contrast between them is unambiguous.

Force supervision is the enabling ingredient. Every variant that predicts the six-dimensional fingertip-force target gains force modulation, and every variant without it stays near zero under the gentle condition: the tactile-image, marker-field, and force-input variants (Img, Field, and Force E) all reach a gentle success rate of 0.01, regardless of which tactile representation they receive. Force supervision alone, with no tactile stream at all (FS), already reaches 0.82 firm and 0.45 gentle, on par with the best tactile-plus-supervision variants on the firm condition.

Impact of tactile input. Without force supervision, adding a tactile image or marker field raises firm-force success only to around 40%, and gentle interaction still fails. The value of touch is realized through force supervision: with supervision enabled, the marker field gives the best firm–gentle trade-off, and most other variants improve on the supervision-only baseline under both conditions. The direction of the contribution is consistent, but its magnitude varies across variants rather than yielding a uniform gain: Force D+FS performs worse than the supervision-only baseline under the gentle condition, and the variation across models exceeds the average gain over supervision alone. This is consistent with the fingertip-force decomposition in Section 3.2: the supervision target is the decomposed contact force, whose grip component follows from the commanded aperture and contact normal, whereas the shear and moment components that separate grip from applied interaction are precisely what the marker displacement field observes. Tactile sensing therefore provides

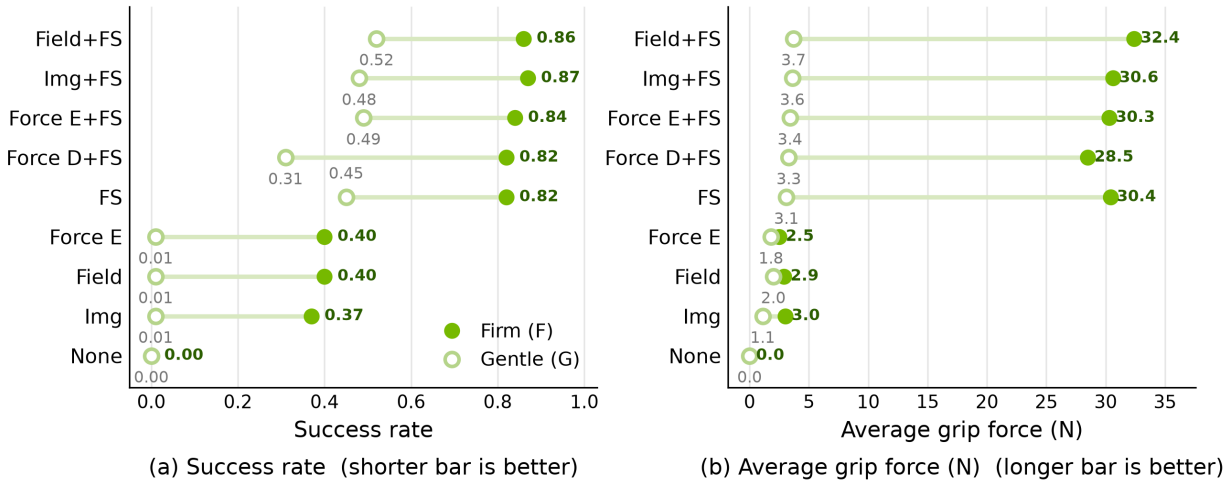


Figure 8: **Ablation study on tactile modalities.** (a) Success rates under firm (F) and gentle (G) force, where a shorter bar (a smaller firm–gentle gap) is better; (b) average grip force (N) under firm and gentle force, where a longer bar (a larger firm–gentle separation) is better. Solid dots mark firm, hollow dots gentle. None = no tactile input; Img = tactile image input; Field = marker displacement field input, i.e. the spatial marker field; Force E = six-dimensional force input via MLP encoder; Force D = six-dimensional force input via decoder; FS = force supervision loss enabled.

observations that make the supervision target resolvable at the contact and allow it to influence the final force and action outputs.

4.4. From Single-Source to Multi-Source Data

We use the data-replay pipeline introduced by Tabero to obtain, from single-distribution real-robot data, simulated Taco data with “firm” and “gentle” force distributions, and evaluate the consistency between the two domains. To complement task-level validation, we measure the consistency along three channels: visual images, tactile images, and action labels. Visual consistency is quantified by the CLIP similarity Radford et al. (2021) of the third-person and wrist camera views; tactile consistency by the structural similarity (SSIM) and the mean absolute error (MAE) between simulated and real tactile images; and action consistency by the MAE of the action labels. Table 3 reports these metrics for three firm-force tasks. The results show that the wrist and third-person camera views are well aligned (both CLIP similarities above 0.75); the tactile signals further agree in SSIM and MAE, and the actions are also well aligned, demonstrating the data reliability of the Taco replay pipeline. Figure 9 shows this alignment qualitatively for one task, pairing the real run with the simulated replay across the third-person, wrist, and tactile views. UMI data further extend the data sources: because UMI data cost less than teleoperation, we can collect data over a larger spatial randomization range. Appendix C.5 shows how the expanded spatial range differs from the original teleoperated data in trajectories and distributions, and Section 4.8 further evaluates the synergy of multi-source data.

4.5. Policy-Level Zero-Shot Sim2Real Transfer

We test whether Taco supports policy-level zero-shot transfer of language-conditioned force levels from simulation to the physical robot. Each policy is initialized from a pretrained VLA and receives task-specific adaptation using simulated data only; after Real2Sim calibration, the same adapted policy, language conditions, and controller are executed in the aligned simulation (Sim) and directly on the physical robot (Sim2Real), without real-robot task demonstrations, policy fine-tuning, or test-time adaptation. The force regimes are changed

Table 3: Sim–real data consistency across visual, tactile, and action channels. All three channels align on every task, indicating that the replay pipeline reproduces the real data distribution closely enough to substitute for real demonstrations on these tasks.

	Third-person CLIP	Wrist CLIP	Tactile SSIM	Tactile MAE	Action MAE
Firm Task 1	0.781	0.770	0.941	0.0160	0.0708
Firm Task 2	0.785	0.754	0.931	0.0182	0.0685
Firm Task 3	0.794	0.766	0.932	0.0196	0.0733

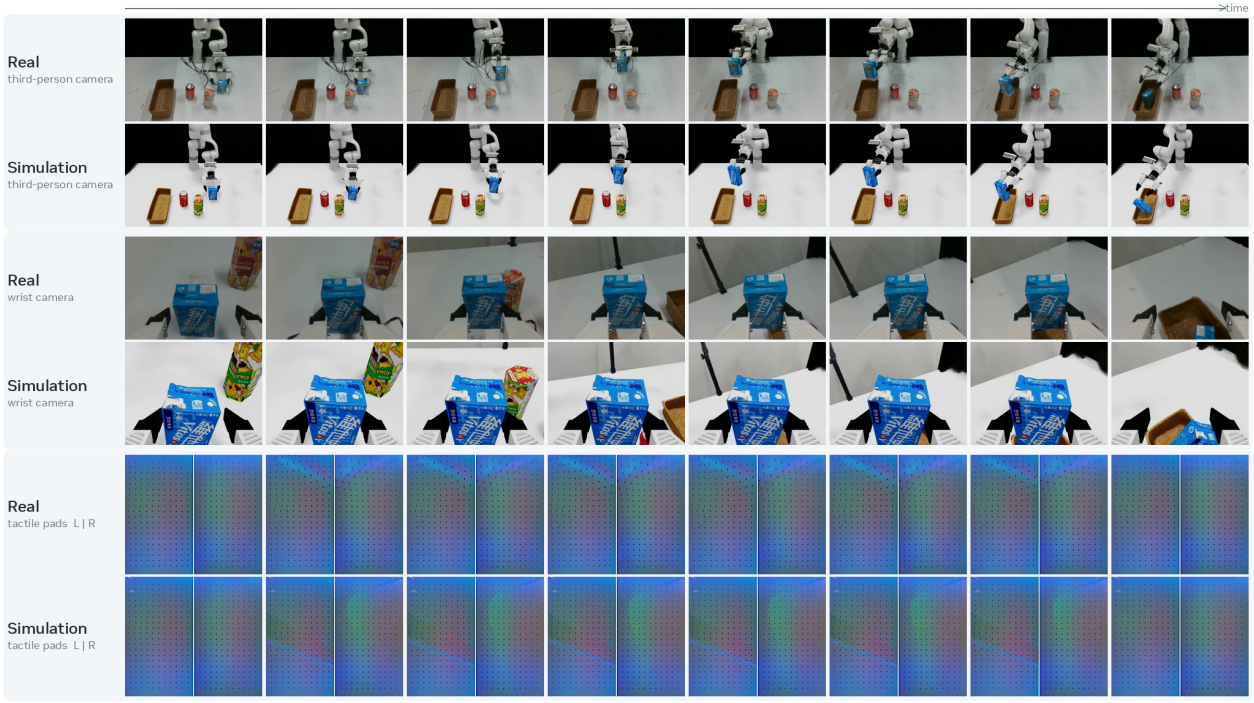


Figure 9: **Real-versus-simulated milk-carton grasping and placement.** Eight time-aligned observations show the real and simulated third-person views, wrist-camera views, and left/right tactile pads as the robot grasps and places a milk carton.

during simulation data generation, while the robot evaluation measures whether the corresponding force level is reproduced. Real tactile observations used for simulator and observation-mapping calibration are separate from policy optimization. Evaluation uses frozen calibrated assets and controller settings, with evaluation scenes and random seeds separated from training. Following the paired evaluation protocol of PolaRiS Jain et al. (2026), the two evaluations share the same task set, language instructions, controller settings, and termination criteria. For every policy we report the mean success rate (Mean SR), averaged over the firm (F) and gentle (G) conditions and over the three tasks, and the average grip force the policy predicts. The reference is the human teleoperation label: for task t and force level $x \in \{\mathrm{F}, \mathrm{G}\}$, the teleoperation demonstrations define $\mathrm{AG}_{x,t}^{\mathrm{teleop}}$, the average grip force the operator applies under the same instruction. We define the signed prediction deviation

$$\Delta_{x,t} = \mathrm{AG}_{x,t}^{\mathrm{pre}} - \mathrm{AG}_{x,t}^{\mathrm{teleop}}, \tag{13}$$

so that $\Delta > 0$ means the policy predicts a firmer grasp than the human label and $\Delta < 0$ a gentler one. Every deviation in this section uses the predicted force *pre*, not the measured force *meas*: the question is whether the policy reproduces the human force label, whereas measurement and controller tracking are evaluated separately in Section 4.6. Each model therefore contributes six deviations per domain, one for each force level

and task, and Table 4 reports the mean of their absolute values, $\bar{\Delta}$, for each domain.

Without tactile input or tactile supervision, the vision-only variants did not solve the evaluated tasks in either domain (Mean SR 0) and provide no force reading. Among the tactile policies, Force D+FS attains the highest Mean SR, followed by Field+FS ($\pi_{0.5}$, no state) and Force E+FS; Field+FS (π_0) is lowest because it fails almost all Vitasoy episodes.

Every tactile model has a mean absolute deviation within 1.4 N, indicating predictions close to the human labels.

Table 4: Predicted grip force against the human teleoperation label. Mean SR (%) averages the firm (F) and gentle (G) success rates over the three tasks. $\bar{\Delta}$ is the mean absolute deviation $|\mathrm{AG}^{\mathrm{pre}} - \mathrm{AG}^{\mathrm{teleop}}|$ over the three tasks and both force levels, in N; larger $\bar{\Delta}$ means the predicted force departs further from the human label. Sim evaluates the adapted policy in the aligned simulation; Sim2Real evaluates it on the physical robot. A dash marks a policy with no force reading; None and Img fail both conditions in both domains. For Field+FS (π_0) on the robot, the Vitasoy gentle condition has no completed episode, so its $\bar{\Delta}$ averages the five remaining cells. Only AG is used; MG is omitted.

	Sim2Real		Sim	
Model	Mean SR (%) ↑	$\bar{\Delta}$ (N) ↓	Mean SR (%) ↑	$\bar{\Delta}$ (N) ↓
None ($\pi_{0.5}$)	0	–	0	–
Img (π_0, w=0)	0	–	0	–
Force D+FS (π_0)	98.3	3.63	99.0	0.62
Force E+FS (π_0)	92.5	1.35	98.0	0.94
Field+FS (π_0)	67.5	0.90	69.3	0.47
Field+FS ($\pi_{0.5}$, state)	82.5	4.12	89.0	0.78
Field+FS ($\pi_{0.5}$, no state)	96.7	1.14	96.7	1.38
Img+FS ($\pi_{0.5}$, no state)	88.3	1.12	95.7	1.11

Table Notes: $\bar{\Delta}$ uses the predicted force *pre*, not the measured force, and averages absolute deviations, so positive and negative firm/gentle errors do not cancel. Simulation AG follows Eq. (2); robot AG uses the two-finger sum described in Section 4. The remaining Img+FS ($\pi_{0.5}$) variants are evaluated in simulation only and reported under the force-separation metric in Appendix C.3.

4.6. Crossing the Sim2Real Gap

Following SIMPLER and PolaRiS Jain et al. (2026); Li et al. (2025b), we quantify the faithfulness of the simulated evaluation with the Pearson correlation and the Mean Maximum Rank Violation (MMRV) between the Sim2Real and Sim scores, computed across the eight policy variants evaluated in both domains: the seven tactile variants with paired Sim and Sim2Real scores, together with the $\pi_{0.5}$ baseline under the position-controlled gripper system. The suite spans two pretrained VLA backbones and several tactile input/force-supervision configurations, so this is a cross-model consistency test rather than a single-checkpoint demonstration. We use the success-only score, Mean SR. Fig. 10 reports $r = 0.998$ and MMRV$= 0.010$: the success ordering transfers almost perfectly.

Fig. 10 also places this agreement alongside recent sim-to-real evaluations. PolaRiS (Jain et al., 2026) reports $r = 0.83$, and its published comparison reports an MMRV of 0.23 for Ctrl-World (Guo et al., 2026). OpenViTac (Wu et al., 2026) reports $r = 0.967$ and MMRV$= 0.006$ across seven methods and eight shared tasks; SimFoundry (Ranawaka et al., 2026) reports $r = 0.911$ and MMRV$= 0.018$ across seven tasks and five policy families; SceneRig (Luo et al., 2026) reports $r = 0.92$ and MMRV$= 0.04$ across ten policy–task pairs. These values use different task suites, policy pools, and scoring protocols, so they are contextual references rather than a direct benchmark.

Table 5: Measured grip force in simulation and on the robot for the four policies with the highest task success rates. S2R = Sim2Real. F SR/G SR = firm/gentle success rate (%); F AG/G AG = measured average grip force (N); Gap = Sim − Sim2Real, so a positive value means the simulation measures a firmer grasp than the robot. The Mean row averages each column over the three tasks. Simulation AG follows Eq. (2); robot AG uses the two-finger sum described in Section 4.

		Sim				Sim2Real				Gap (Sim−S2R)	
Model	Task	F SR	G SR	F AG	G AG	F SR	G SR	F AG	G AG	F AG	G AG
Force D+FS (π_0)	Vitasoy	98.0	96.0	25.29	9.00	100.0	100.0	19.66	13.23	+5.63	-4.23
	Coca-Cola	100.0	100.0	7.73	4.54	100.0	100.0	7.95	6.42	-0.22	-1.88
	Cookie	100.0	100.0	24.53	7.46	90.0	100.0	6.87	6.27	+17.66	+1.19
	Mean	99.3	98.7	19.18	7.00	96.7	100.0	11.49	8.64	+7.69	-1.64
Force E+FS (π_0)	Vitasoy	98.0	94.0	26.42	10.27	100.0	90.0	10.81	5.72	+15.61	+4.55
	Coca-Cola	96.0	100.0	7.80	4.70	100.0	100.0	10.03	8.25	-2.23	-3.55
	Cookie	100.0	100.0	25.13	8.22	70.0	95.0	7.55	3.08	+17.58	+5.14
	Mean	98.0	98.0	19.78	7.73	90.0	95.0	9.46	5.68	+10.32	+2.05
Field+FS ($\pi_{0.5}$, NS)	Vitasoy	98.0	94.0	27.23	10.05	85.0	95.0	20.14	10.29	+7.09	-0.24
	Coca-Cola	100.0	100.0	7.80	4.74	100.0	100.0	8.12	7.47	-0.32	-2.73
	Cookie	94.0	94.0	25.58	8.43	100.0	100.0	18.56	10.37	+7.02	-1.94
	Mean	97.3	96.0	20.20	7.74	95.0	98.3	15.60	9.38	**+4.60**	**-1.64**
Img+FS ($\pi_{0.5}$, NS)	Vitasoy	88.0	88.0	26.47	10.69	90.0	80.0	13.05	10.60	+13.42	+0.09
	Coca-Cola	100.0	100.0	7.78	5.10	100.0	100.0	10.46	7.01	-2.68	-1.91
	Cookie	100.0	98.0	25.36	8.48	80.0	80.0	18.90	10.59	+6.46	-2.11
	Mean	96.0	95.3	19.87	8.09	90.0	86.7	14.14	9.40	**+5.73**	**-1.31**

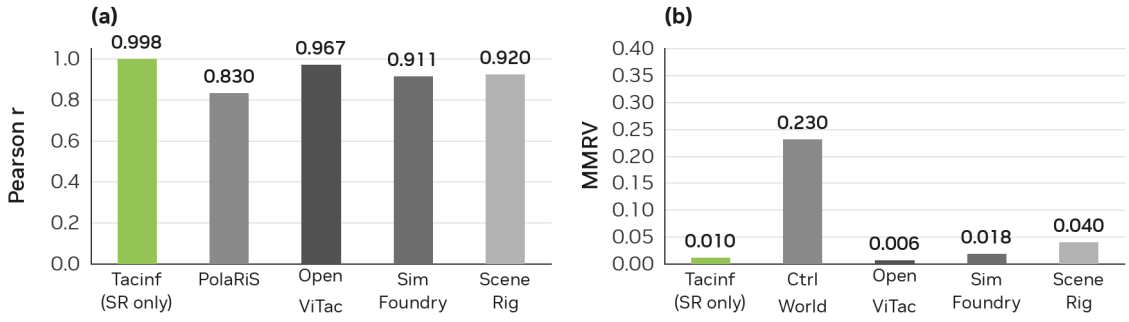


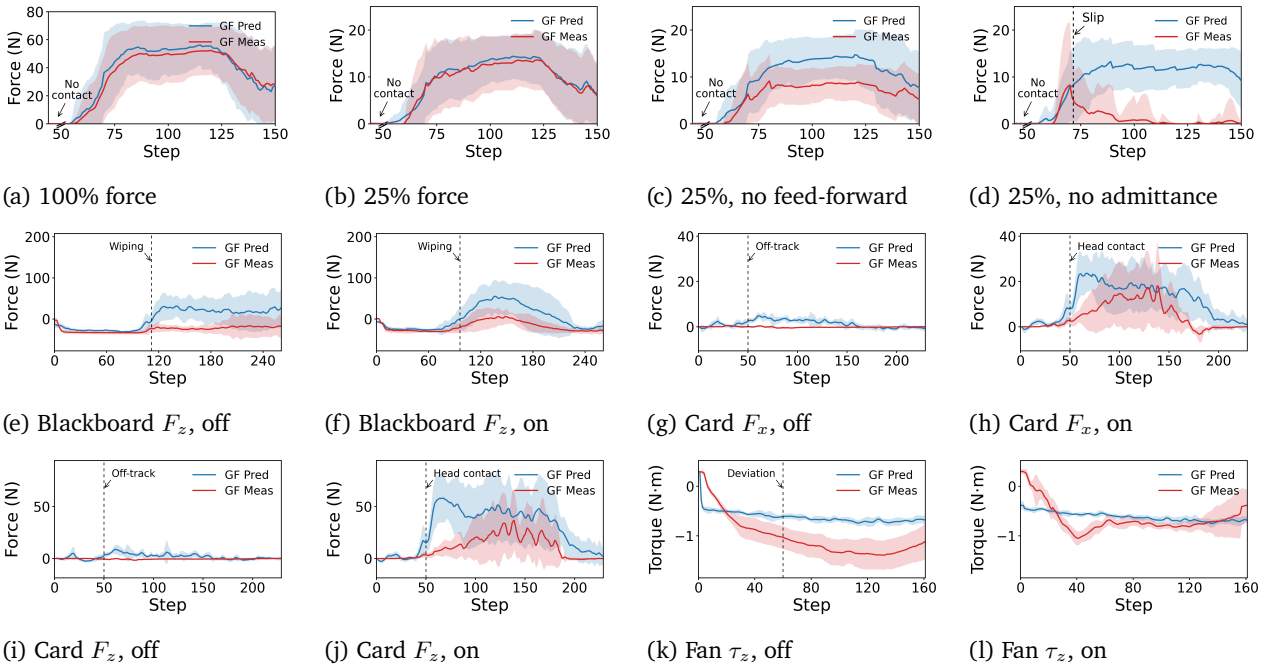
Figure 10: **Cross-domain agreement of simulated evaluations.** (a) Pearson correlation between simulated and real-robot scores, higher is better; (b) Mean Maximum Rank Violation (MMRV), lower is better. Across eight policy variants, the success-only score gives $r = 0.998$ and MMRV$= 0.010$. Literature bars: PolaRiS $r = 0.83$ (Jain et al., 2026), Ctrl-World MMRV$= 0.23$ (Guo et al., 2026), OpenViTac $r = 0.967$/MMRV$= 0.006$ (Wu et al., 2026), SimFoundry $r = 0.911$/MMRV$= 0.018$ (Ranawaka et al., 2026), and SceneRig $r = 0.92$/MMRV$= 0.04$ (Luo et al., 2026). Each literature value is reported by the corresponding paper on its own benchmark; the task suites, policy pools, and scoring protocols differ, so these bars are contextual references, not directly comparable benchmark results.

Table 5 turns from predicted to measured force for the four policies with the highest task success rates, reporting the measured AG in both domains and the per-task gap $\mathrm{AG}^{\mathrm{meas}}_{\mathrm{Sim}} - \mathrm{AG}^{\mathrm{meas}}_{\mathrm{S2R}}$; a positive gap means that the simulation measures a firmer grasp than the robot. Although the field- and image-based $\pi_{0.5}$ NS policies trail the force-based models in success, their mean sim-to-real gaps are smaller. The measured values also differ

from the predicted values; this difference comes from the gripper controller parameters rather than from model capability.

4.7. Execution of Force-Position Control

To isolate the contribution of the low-level controller from the high-level policy, we evaluate on a mixture of the Tabero **Dataset A** and Taco special-task data, with 30 demonstrations per task collected by teleoperation. The hyperparameters of our controller are presented in Appendix A.1. In Tabero, we use the best-performing Field+FS model and conduct four ablation studies on the gripper controller: (a) full force with hybrid control, (b) reduced force with hybrid control, (c) reduced force without feedforward term, and (d) reduced force without admittance component. Panels (a)–(d) of Fig. 11 demonstrate that both feedforward and admittance terms are essential for accurate force tracking; without closed-loop control, objects slip, causing task failure.



(a) 100% force (b) 25% force (c) 25%, no feed-forward (d) 25%, no admittance

(e) Blackboard F_z, off (f) Blackboard F_z, on (g) Card F_x, off (h) Card F_x, on

(i) Card F_z, off (j) Card F_z, on (k) Fan τ_z, off (l) Fan τ_z, on

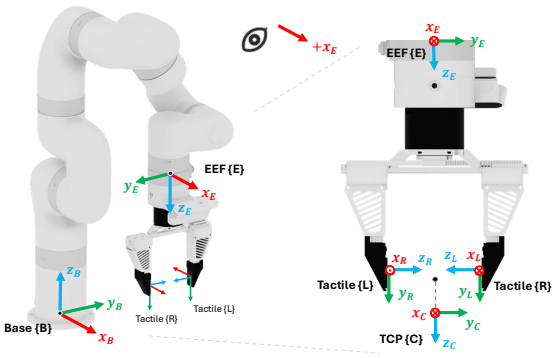


Figure 11: **Gripper force control.** Predicted (blue) and measured (red) forces; shaded bands show run-to-run standard deviation. (a)–(d) Simulated controller ablations; (e)–(l) real-robot blackboard wiping, card swiping, and fan turning with the loop off and on. GF denotes gripper force, and Slip marks object loss. The coordinate schematic identifies the robot base, end-effector, TCP, and tactile-sensor frames.

We train and test the Field+FS ($\pi_{0.5}$, no state) model that uses the tactile marker field in the preceding evaluation on new real-robot tasks. The new tasks are the contact-rich blackboard wiping, card swiping, and knob turning; Appendix B.2 defines their task-level success-score criteria, and Fig. 12 presents the test results judged by these criteria together with illustrative process frames. Blackboard wiping requires controlling the downward pressure along the z-axis, card swiping requires maintaining a stable downward force along z and a normal applied force along x, and knob turning requires controlling the torsional torque. Panels (e)–(l) of Fig. 11 show the force-tracking performance for the three tasks, one run each with the closed loop on and off. With the closed-loop controller on, the measured TCP force stays closer to the predicted command, most clearly where the command changes fastest, and the overall task success score is higher; with the controller off, the measured force poorly follows the prediction and the success score drops: blackboard wiping falls to zero and card swiping falls from 7 to 3 points, while knob turning is less affected because the gripper behavior is relatively fixed and the contact is biased to one side, although panels (k) and (l) of Fig. 11 still show the core difference.

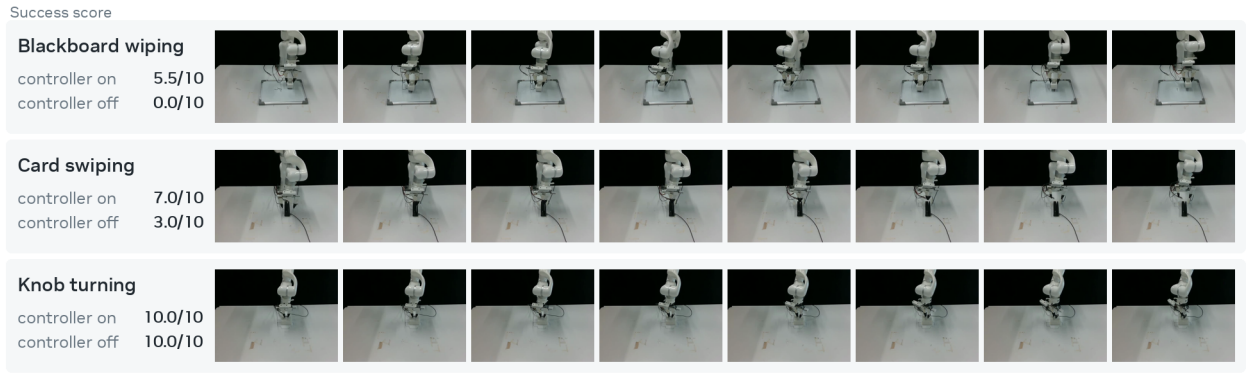


Figure 12: **Real-robot task expansion.** One row per task shows eight keyframes in time order. Scores at left are out of 10, computed over 10 trials for Field+FS ($\pi_{0.5}$, no state), with the controller on and off.

4.8. Data Synergy

We evaluate whether the augmentation and joint post-training loop described in Section 3.4.3 can realize a data synergy effect, i.e., whether data of different costs and volumes can each play to their characteristics and endow the policy with multiple capabilities. We set teleoperation data to fixed position and fixed object, UMI data to random position and fixed object, and simulation data, based on teleoperation replay, to consistent fixed position and random object. In terms of cost, teleoperation data take longer to collect than UMI data, and simulation data are the cheapest. Each share of data consists of 30 demonstrations for each of the six Taco base tasks, and each policy is trained directly on mixtures of different numbers of shares using the same coordinate system, observation, action, timestep, and normalization interfaces. Table 6 reports six training mixes—only real, only umi, real+umi (1:1), sim+umi (5:1), real+sim (1:5), and real+sim+umi (1:5:1)—on the six original tasks, evaluated under position randomization and position+object randomization. The randomized task setup is summarized in Appendix B.3. All entries are firm/gentle success rates and the average grip force predicted by the policy on held-out real-robot scenes.

Figure 13 shows examples of the simulation training objects and real-robot test objects used to evaluate object generalization.

Key findings.

A small amount of UMI data alone cannot complete the tasks. The UMI-only mix attained 0% firm success in both settings. Its contribution appears only in combination with real or simulated trajectories, consistent



(a) Simulation training objects

(b) Real-robot test objects

Figure 13: **Objects used for mixed-data evaluation.** Simulation training assets (left) and physical test objects (right).

Table 6: Real–simulation–UMI co-training results under different training mixtures on the six original tasks with background disturbance. Each mix is evaluated under two randomization settings: position randomization, and position+object randomization. Training mixes are given as relative sample volumes; the UMI share is Tacumi demonstrations with object positions randomized within a range. F SR/G SR = success rates under firm/gentle force, reported as fractions; F AG/G AG = predicted average grip force (N) commanded by the policy, before real-robot tracking. A dash marks a condition with no recorded value. Bold and underlined values indicate the best and second-best results, respectively.

	Position randomized				Position + object randomized			
Training Mix	F SR	G SR	F AG (N)	G AG (N)	F SR	G SR	F AG (N)	G AG (N)
only real	0.300	0.333	13.52	7.81	0.200	0.200	13.71	9.17
only umi	0.000	–	–	–	0.000	–	–	–
real+umi (1:1)	**0.833**	0.700	17.96	8.44	0.100	0.100	13.36	10.60
sim+umi (5:1)	<u>0.700</u>	<u>0.633</u>	21.47	9.15	<u>0.400</u>	<u>0.400</u>	24.66	9.34
real+sim (1:5)	0.433	0.367	21.57	9.73	0.200	0.400	25.84	8.27
real+sim+umi (1:5:1)	0.567	**0.833**	22.04	10.47	**0.600**	**0.600**	24.08	13.32

with the role of UMI as spatial augmentation rather than a stand-alone training source. Under position randomization, every multi-source mixture improved on the real-only baseline. Mean success was highest for real+umi, whose data most closely match the test scenarios, followed by real+sim+umi. Adding UMI to real data raised firm success from 30.0% to 83.3% and gentle success from 33.3% to 70.0%. Under position+object randomization the ordering changed: real+umi and real-only both showed substantial drops, indicating that the distribution shift makes out-of-distribution tasks difficult, whereas sim+umi reached 40.0%, real+sim 30.0%, and real+sim+umi 60.0%. The three-source mixture improved over real-only by 38.3 and 40.0 percentage points in the two settings, the only multi-source mix with a gain above 30 points in both; its gain was largest under object randomization and second only to real+umi under position randomization. The contrast identifies the division of labor: UMI demonstrations supply object-position and approach variation, simulation supplies object-appearance and background variation, and real trajectories anchor the sensor and actuator statistics. Of course, the increase in data volume is also a factor that cannot be ignored, but the setup and results of real+sim show that the capability gain does not rely solely on more data; the characteristics and diversity of the data themselves matter more.

Predicted force separation is positive for every mixture and setting with a recorded force statistic, and every mixture that produced a force statistic separates firm and gentle in the predicted target; however, we attribute the increase in separation mainly to the larger data volume, especially after large amounts of simulation data are added, where the force prediction moves closer to the original teleoperation labels.

Taken together, the results support a clear conclusion. The three data sources form a data pyramid that balances data efficiency and collection effort: the three-source mixture improves mean success on held-out real-robot tasks with position and out-of-distribution object randomization from 20.0% to 60.0%. In this low-data regime, UMI-only training fails and UMI data require teleoperated data for calibration, while sim+umi training without teleoperated data also performs well under randomized conditions. Overall, co-training improves task competence with limited data.

5. Conclusions

We present Tacinf as a modular tactile foundation model infrastructure for force-aware manipulation. The policy interface uses task language, vision, and touch to predict actions and fingertip-force targets; an explicit language-conditioned force level can be added when the task specifies how firmly to interact. Ablation experiments demonstrate the roles of modular tactile inputs and force supervision, as well as the force-tracking capability of the controller. After simulation-only policy adaptation, the policy transfers to real-robot evaluation without real-robot policy fine-tuning. Across eight paired policy variants evaluated on six tasks, simulation preserves the real-robot success ordering almost perfectly while maintaining a small measured-force sim-to-real gap. Co-training experiments show that teleoperation, simulation, and UMI data form a data pyramid balancing data efficiency and collection effort: the three-source mixture improves mean success on held-out real-robot tasks with position and out-of-distribution object randomization from 20.0% to 60.0%, and this gain does not follow directly from the increase in data volume. In this low-data regime, UMI-only training fails and UMI data require teleoperated data for calibration, while sim+umi training without teleoperated data also performs well under randomized conditions.

Limitations. Because of resource constraints, the data and task scale in this study is limited, and we do not conduct large-scale training. Future work will focus on further scaling up on the data pyramid and extending to different robot embodiments and bimanual systems.

A. Training and Interface Contract

Detailed training hyperparameters and model configurations will be released with the code. Table 7 lists the tactile input variants used in the ablations. The Xense values described in this paper refer to the policy-visible observation interface rather than a mechanically verified sensor.

Table 7: Tactile input variants used in the ablations. "Prefix" and "suffix" denote the injection location in the backbone; "FS" denotes force supervision.

Variant	Visual streams	Tactile stream	Adapter	FS
Img+FS	3	–	–	yes
Img	3	–	–	no
Force D+FS	2	suffix	MLP	yes
Force E	2	prefix	MLP	no
Force E+FS	2	prefix	MLP	yes
Field+FS	2	prefix	TCN	yes
Field	2	prefix	TCN	no
FS	2	–	–	yes
None	2	–	–	no

Interface items to release.

The public data contract should specify, for each stream, units, frame convention, timestamp policy, normalization statistics, and missing-value handling. The release manifest should specify camera frame rate, RGB-D resolution, and the normalization-file path. The policy and the tactile closed-loop controller both run at 20 Hz; the policy executes asynchronously through temporal ensembling, with an inference latency of roughly 150–200 ms.

A.1. Controller Settings

The two-loop controller tracks the policy's pose and fingertip-force targets. The values in Table 8 are controller settings for the Hybrid+Tactile configuration and are not claimed to be hardware-independent defaults.

Table 8: Force–position hybrid-controller settings.

Parameter	Value
Grip feed-forward gain / deadzone	0.6 / 1.0
Cartesian position gains (x, y, z)	$(-10^{-4}, -10^{-4}, -10^{-4})$
Grip proportional gain / deadzone	0.0008 / 0.25
Force filter coefficient α	0.2
IK command / solver	pose / DLS
IK scale / position offset	1.0 / $[0, 0, 0.1034]$

B. Task Protocol

B.1. Language-Conditioned Force Protocol

For each base instruction, force-related adverbs are inserted either before or after the verb phrase and the insertion position is randomized. The in-domain pairs are "firmly/tightly" for the high-force condition and

"gently/softly" for the low-force condition. Unseen adverbs such as "lightly" and "forcefully" are reserved for the generalization test and never enter training. Table 9 gives example templates.

Table 9: Example instruction templates for force-language conditioning.

Template	Instruction
Prefix-high	*firmly pick up the alphabet soup and place it in the basket*
Suffix-high	*pick up the alphabet soup and place it in the basket tightly*
Prefix-low	*softly pick up the alphabet soup and place it in the basket*
Suffix-low	*pick up the alphabet soup and place it in the basket gently*

The Tabero subset contains the nine base tasks listed in Table 10. The force adverbs are applied to every task using the protocol above.

Table 10: Base tasks in the Tabero subset.

ID	Task description
0	pick up the alphabet soup and place it in the basket
1	pick up the cream cheese and place it in the basket
2	pick up the salad dressing and place it in the basket
3	pick up the bbq sauce and place it in the basket
4	pick up the tomato sauce and place it in the basket
5	pick up the butter and place it in the basket
6	pick up the milk and place it in the basket
7	pick up the chocolate pudding and place it in the basket
8	pick up the orange juice and place it in the basket

B.2. Task-Specific Success Criteria

Blackboard Wiping.

The score is computed from the amount of handwriting remaining after the wipe, compared against the demonstration data. If the residue reaches the level observed in human demonstrations, the score is 1; if minor residue remains (less than one third of the handwriting length), the score is 0.5; if significant residue remains, the score is 0.

Card Swiping.

A trial is considered successful when the card traverses the predefined path in the intended direction while remaining functionally engaged with the slot and correctly oriented. The card need not remain in contact with the bottom of the slot, but its lower edge should stay within the vertical range observed in human demonstrations. The swipe should proceed continuously, without sustained pauses or substantial reverse motion.

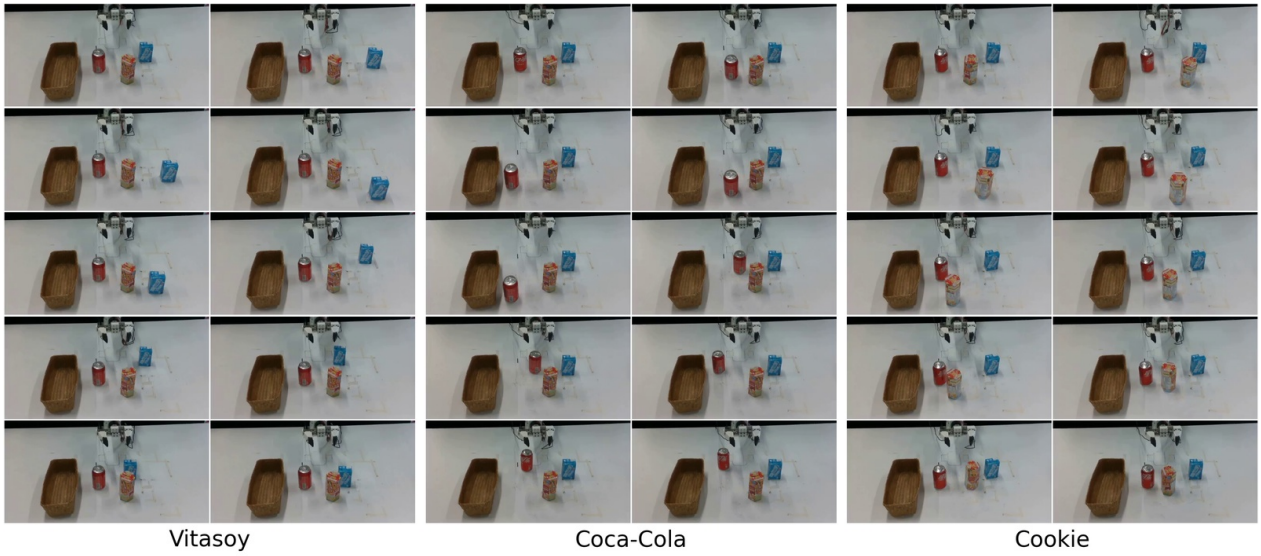
Knob Turning.

A trial is considered successful when the gripper maintains stable bilateral contact with the knob and continuously drives it with approximately steady rotational motion. Knob rotation should remain coupled with the end-effector motion throughout the manipulation. Fingertip slip or gripper rotation without corresponding knob motion is considered a failure. Rotation produced primarily through unilateral tangential pushing, incidental

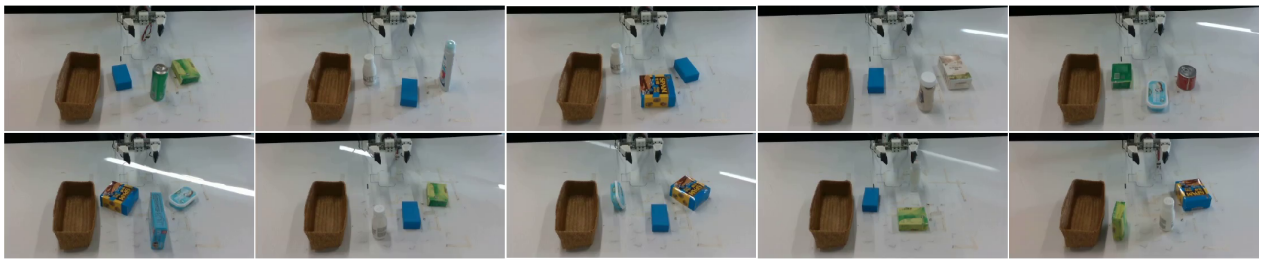
collision, or sliding against the knob is likewise treated as failures. Minor variations in gripper position and orientation are acceptable if they remain within the demonstration-derived range.

B.3. Position and Object Randomization Protocol

The real-robot evaluation in Section 4.8 comprises six Taco tasks: three with randomized initial object positions and three with both randomized initial positions and object instances (Fig. 14).



(a) Position randomization



(b) Position and object randomization

Figure 14: **Randomized task configurations.** (a) Position variation with the original Vitasoy, Coca-Cola, and cookie objects. (b) Joint position and object variation. The basket remains the placement goal in both settings.

C. Supplementary Evaluation

C.1. Retention Summary

We define retention as the fraction of replay episodes that complete the task successfully, $R = N_s/N_t$, where N_s is the number of episodes that complete the task and N_t is the number of replayed episodes.

C.2. Semantic-Force Generalization

Table 11 reports the intended format for the in-domain and out-of-domain linguistic-adverb test. The draft values are retained for the current snapshot; they must be regenerated if the dataset, checkpoint, or controller changes.

Table 11: Semantic-force understanding on Dataset B. Force values are in N.

Adverb	SR	AG	AA	MG	MA
firmly, tightly	0.86	32.4	1.5	62.7	6.9
gently, softly	0.52	3.7	0.4	11.4	5.1
lightly (OOD)	0.61	14.5	0.8	40.8	7.6
forcefully (OOD)	0.64	18.1	0.9	49.7	6.3

The test is interpreted as a language-to-force trend, not as a calibration of absolute force. Report the number of episodes, random seeds, and confidence intervals in the release manifest; these fields are currently *TBD*.

C.3. Img+FS Variants

Table 12 reports the remaining Img+FS ($\pi_{0.5}$) variants under simulation.

Table 12: Additional Img+FS ($\pi_{0.5}$) variants under simulation. The last column is the simulated firm–gentle force separation F AG − G AG, not the prediction deviation used in Section 4.5.

Model	Sim SR (%)	F AG − G AG (N)
Img+FS (state)	86.35	16.21
Img+FS (no state)	90.65	13.55
Img+FS (state, $H = 1$)	91.70	16.74
Img+FS (no state, $H = 1$)	95.65	15.39

C.4. Sensor Measurement Validation

Figure 15 compares the Xense SDK-estimated resultant force components across five grasp configurations, some of which exhibit left–right asymmetry. Grip force (AG) is computed according to Eq. (3).

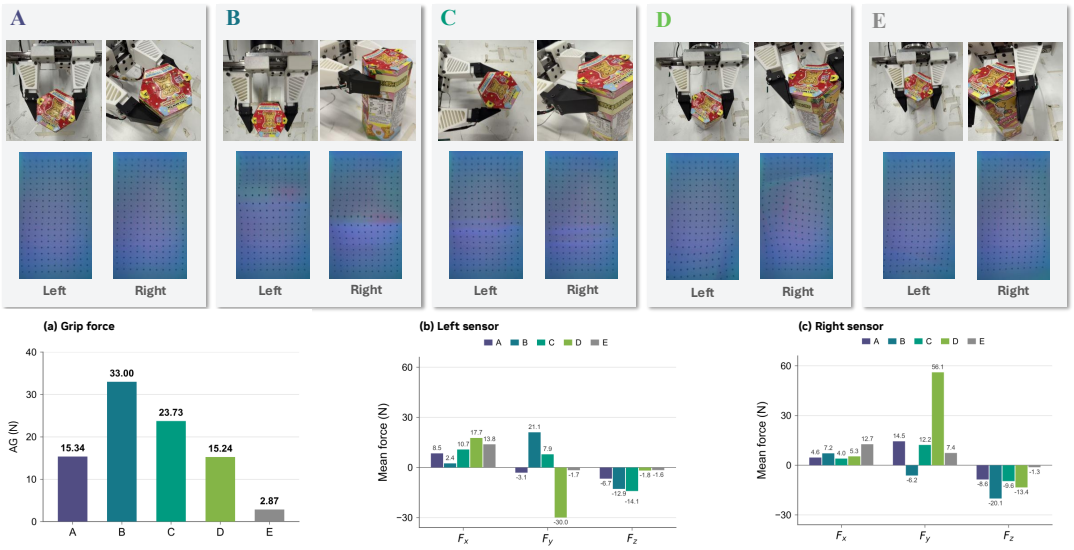


Figure 15: **Fingertip force measurements.** Top: configurations A–E and paired tactile images. Bottom: (a) grip force (AG), computed according to Eq. (3); (b,c) time-averaged components of the resultant force reported by the left and right tactile sensors.

C.5. UMI Data Distribution Validation

Figure 16 compares UMI and teleoperation trajectories for three manipulation tasks. The comparison shows overlapping task paths and additional spatial variation in the UMI demonstrations.

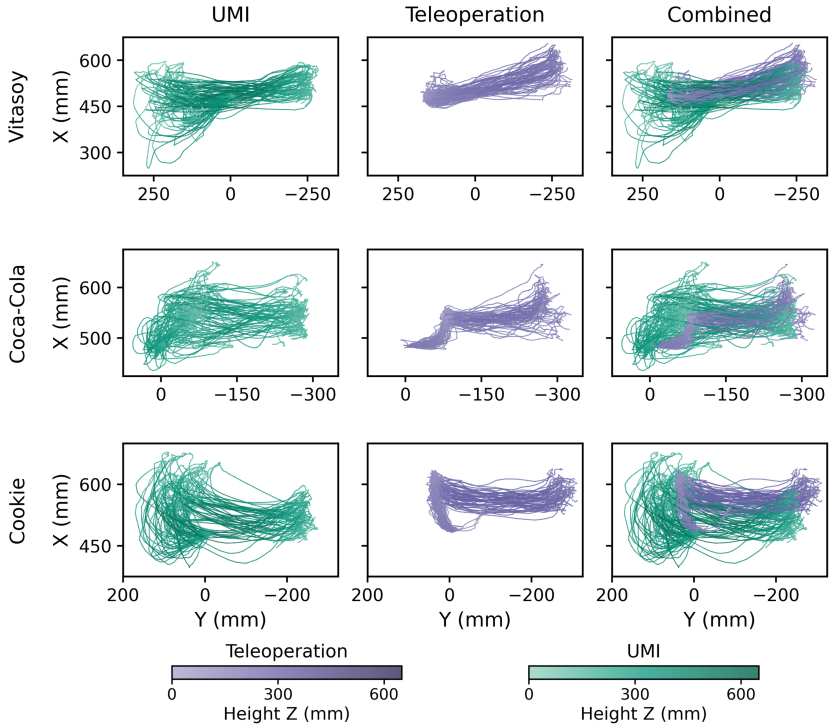


Figure 16: **Spatial coverage of UMI and teleoperation demonstrations.** Rows: Vitasoy, Coca-Cola, and cookie tasks. Columns: UMI, teleoperation, and their overlay. Curves show trajectory projections onto the X–Y plane. Purple denotes teleoperation and green denotes UMI; color shade encodes height Z. All coordinates are in millimetres.

D. Tactile Real2Sim Calibration

We calibrate the tactile observation interface for contact onset, patch persistence, force scale, marker response, and channel and temporal conventions.

D.1. Depth-to-Indentation Mapping

At each tactile update, the camera produces a depth map $D(u, v)$ in metres. We express the same map in millimetres as $Z(u, v) = 1000\, D(u, v)$. Let $d_{\min} = \min_{u,v} D(u, v)$, d_{ref} be the camera-to-gel reference distance, and h_g the effective gel thickness, all in metres. The distance offset δ (metres) and shared indentation depth h_{in} (millimetres) are

$$\delta = \max(0, d_{\min} - d_{\text{ref}})\,, \qquad h_{\text{in}} = \begin{cases} (h_g - \delta)\, 1000, & \delta \le h_g, \\ 0, & \delta > h_g. \end{cases} \tag{14}$$

The estimated indentation is positive when $d_{\min} < d_{\text{ref}} + h_g$ and saturates at $1000 h_g$ for $d_{\min} \le d_{\text{ref}}$. The relative depth field used to compute gel deformation is $Z'(u, v) = Z(u, v) - \min_{u,v} Z(u, v) - h_{\text{in}}$.

D.2. Taxim and FOTS Interface

Taxim Si and Yuan (2022) uses calibrated polynomials to map local gradients of the deformed gel and normalized image positions to RGB intensity changes, which are added to the real no-contact background. Pixels outside the contact mask retain the background. The output uses RGB ordering and no additional shadow term.

FOTS Zhao et al. (2024) uses the same relative depth field Z' and derived contact mask to approximate normal expansion, shear, and planar torsion with Gaussian spatial decay kernels

$$g_k(x, y) = \exp\left[-\lambda_k\left((x - c_x)^2 + (y - c_y)^2\right)\right], \tag{15}$$

where marker positions (x, y) and kernel centers (c_x, c_y) are in image coordinates, and λ_k controls the decay of component k. Contact-centroid changes provide the tangential cue, and relative sensor–object yaw provides the rotational cue. Marker displacements are expressed in sensor image coordinates. With zero indentation, the trajectory is cleared and markers return to their reference grid.

D.3. Calibration and Scope

The calibration bundle contains the real no-contact background, polynomial coefficients fitted for each gradient bin using real sphere-contact images, and the gel reference map. Camera geometry, pad dimensions, marker scale, spatial decay coefficients, and update settings are specified in the released simulator configuration. The release manifest should record its path and commit hash. The virtual camera pose and depth clipping range align the observations and operating depth range rather than specify the physical sensor internals.

Policy observations follow the 20 Hz sampling convention and configured history length H. Force labels for the Taxim–FOTS backend come from Isaac contact sensors and follow Eq. (4); real-robot force measurements use the manufacturer's Xense calibration.

E. Tacinf-Tacumi Calibration and Synchronization

This section describes the spatial calibration and temporal convention of the handheld data collection system in Section 3.3. Spatial calibration maps the tracked Quest controller pose to the gripper center, while a common host time base relates the visual, tactile, pose, and motor-state streams.

E.1. Hand–Eye Calibration

We estimate the rigid transformation from the Quest controller frame to the gripper-center frame using a robot-assisted optical hand–eye calibration procedure. The controller is rigidly mounted on the robot, and 20 paired poses are collected from the Quest tracking system and the robot. The gripper center defines the tool center point (TCP) used for the recorded manipulation trajectory. This controller-to-tool calibration follows the geometric formulation used in controller-based handheld acquisition Li et al. (2025a).

Figure 17 illustrates the calibration setup and the coordinate-frame relationships.

Let W, B, C, and G denote the Quest tracking reference frame, robot base frame, controller frame, and gripper-center frame, respectively. A transformation ${}^{A}\mathbf{T}_{B}$ maps coordinates from frame B to frame A. For calibration pose i, let $\mathbf{Q}_i = {}^{W}\mathbf{T}_{C,i}$ be the tracked controller pose and $\mathbf{P}_i = {}^{B}\mathbf{T}_{G,i}$ be the corresponding robot-side pose expressed at the gripper center. The unknown controller-to-gripper transformation $\mathbf{X} = {}^{C}\mathbf{T}_{G}$ and the fixed reference-frame transformation $\mathbf{Z} = {}^{W}\mathbf{T}_{B}$ satisfy

$$\mathbf{Q}_i\mathbf{X} = \mathbf{Z}\mathbf{P}_i, \qquad i = 1, \ldots, 20. \tag{16}$$

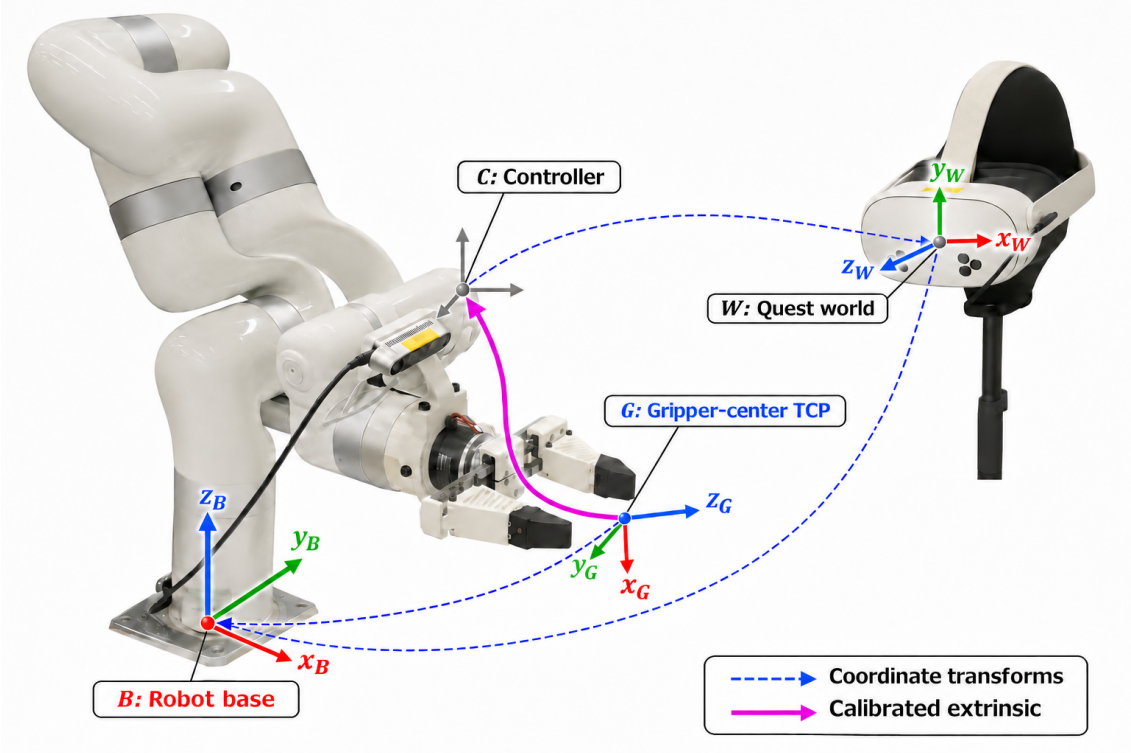


Figure 17: **Hand–eye calibration setup and coordinate frames of Tacinf–Tacumi.** The Quest controller is rigidly mounted with the gripper on an xArm 7. The frames B, G, C, and W denote the robot base, gripper-center TCP, controller, and Quest world, respectively. Blue dashed arrows indicate coordinate-transform relationships; the magenta arrow highlights the fixed gripper-to-controller extrinsic estimated by hand–eye calibration.

Eliminating $\mathbf{Z}$ between two observations yields the relative-motion hand–eye equation

$$\underbrace{\mathbf{Q}_j^{-1}\mathbf{Q}_i}_{\mathbf{A}_{ji}}\mathbf{X} = \mathbf{X}\underbrace{\mathbf{P}_j^{-1}\mathbf{P}_i}_{\mathbf{B}_{ji}}, \tag{17}$$

which has the standard form $\mathbf{AX} = \mathbf{XB}$. We solve for the fixed transformation $\mathbf{X}$ from the collected pose pairs. The notation assumes that the two pose streams are expressed using consistent coordinate conventions and length units.

During handheld collection, the calibrated transformation converts each controller pose to the gripper-center pose as ${}^{W}\mathbf{T}_G(t) = {}^{W}\mathbf{T}_C(t)\mathbf{X}$, matching the TCP transformation in Section 3.3. This mapping accounts for both the translational offset and rotational alignment between the controller and the grasping tool. The calibration remains applicable while their relative mounting geometry is unchanged; changing that geometry requires updating the extrinsic transformation.

E.2. Multimodal Temporal Synchronization

The collection system acquires Quest controller poses, visual images, tactile observations, and gripper motor states at different native rates, as summarized in Table 13. Timestamps for all four streams are generated on the same host computer, providing a common temporal reference for recording the demonstrations.

For modality m, the recorded stream is represented as $\mathcal{D}_m = \{(t_{m,k}, \mathbf{z}_{m,k})\}_k$, where $t_{m,k}$ is the host-generated timestamp and $\mathbf{z}_{m,k}$ denotes the corresponding measurement. Because the sampling rates differ, sample indices

Table 13: Native acquisition rates of Tacinf-Tacumi. All streams use timestamps generated on the same host computer. These rates describe data acquisition rather than the policy execution frequency.

Data stream	Acquisition rate (Hz)
Quest controller pose	72
Visual camera	30
Tactile sensor	100
Gripper motor state	200

do not directly identify simultaneous observations across modalities. The shared timestamp reference provides the basis for associating gripper motion and motor state with visual and tactile observations.

A common host time base establishes timestamp comparability but does not by itself establish simultaneous physical acquisition. Sensor acquisition, transport, and buffering can introduce modality-dependent delays; prior handheld systems explicitly distinguish timestamp alignment from per-modality latency compensation Li et al. (2025a). Accordingly, the common-clock convention described here does not imply a measured synchronization-error bound.

References

Bo Ai, Stephen Tian, Haochen Shi, Yixuan Wang, Cheston Tan, Yunzhu Li, and Jiajun Wu. RoboPack: Learning Tactile-Informed Dynamics Models for Dense Packing. In *Proceedings of Robotics: Science and Systems*, Delft, Netherlands, 2024. doi: 10.15607/rss.2024.xx.130. URL https://doi.org/10.15607/rss.2024.xx.130. 5

Iretiayo Akinola, Jie Xu, Jan Carius, Dieter Fox, and Yashraj Narang. TacSL: A Library for Visuotactile Sensor Simulation and Learning. *IEEE Transactions on Robotics*, 41:2645–2661, 2025. doi: 10.1109/tro.2025.3547267. URL https://doi.org/10.1109/tro.2025.3547267. 3

Manish Bhardwaj. Sensored field oriented control of 3-phase permanent magnet synchronous motors. Technical Report SPRABQ2, Texas Instruments, 2013. URL https://www.ti.com/lit/an/sprabq2/sprabq2.pdf. 13

Jianxin Bi, Kevin Yuchen Ma, Ce Hao, Mike Shou Zheng, and Harold Soh. VLA-Touch: Enhancing Vision-Language-Action Model With Dual-Level Tactile Feedback. *IEEE Robotics and Automation Letters*, 11(7): 8487–8494, 2026. doi: 10.1109/lra.2026.3692345. URL https://doi.org/10.1109/lra.2026.3692345. 5

Kevin Black, Noah Brown, James Darpinian, Karan Dhabalia, Danny Driess, Adnan Esmail, Michael Robert Equi, Chelsea Finn, Niccolo Fusai, Manuel Y. Galliker, Dibya Ghosh, Lachy Groom, Karol Hausman, brian ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Devin LeBlanc, Sergey Levine, Adrian Li-Bell, Mohith Mothukuri, Suraj Nair, Karl Pertsch, Allen Z. Ren, Lucy Xiaoyang Shi, Laura Smith, Jost Tobias Springenberg, Kyle Stachowicz, James Tanner, Quan Vuong, Homer Walke, Anna Walling, Haohuan Wang, Lili Yu, and Ury Zhilinsky. $\pi_{0.5}$: A Vision-Language-Action Model with Open-World Generalization. In Joseph Lim, Shuran Song, and Hae-Won Park, editors, *Proceedings of The 9th Conference on Robot Learning*, volume 305 of *Proceedings of Machine Learning Research*, pages 17–40. PMLR, 2025a. URL https://proceedings.mlr.press/v305/black25a.html. 5, 8

Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Robert Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Sergey Levine, Adrian Li-Bell, Mohith Mothukuri, Suraj Nair, Karl Pertsch, Lucy Xiaoyang Shi, Laura Smith, James Tanner, Quan Vuong, Anna Walling, Haohuan Wang, and Ury Zhilinsky. π_0: A Vision-Language-Action Flow Model for General Robot Control. In *Proceedings of Robotics: Science and Systems*, LosAngeles, CA, USA, 2025b. doi: 10.15607/rss.2025.xxi.010. URL https://doi.org/10.15607/rss.2025.xxi.010. 5, 8, 11

Baijun Chen, Weijie Wan, Tianxing Chen, Xianda Guo, Congsheng Xu, Yuanyang Qi, Haojie Zhang, Longyan Wu, Tianling Xu, Zixuan Li, Yizhe Wu, Rui Li, Xiaokang Yang, Ping Luo, Wei Sui, and Yao Mu. UniVTAC: A Unified Simulation Platform for Visuo-Tactile Manipulation Data Generation, Learning, and Benchmarking. *CoRR*, abs/2602.10093, 2026. URL https://arxiv.org/abs/2602.10093. 3

Tianxing Chen, Zanxin Chen, Baijun Chen, Zijian Cai, Yibin Liu, Zixuan Li, Qiwei Liang, Xianliang Lin, Yiheng Ge, Zhenyu Gu, Weiliang Deng, Yubin Guo, Tian Nian, Xuanbing Xie, Qiangyu Chen, Kailun Su, Tianling Xu, Guodong Liu, Mengkang Hu, Huan-ang Gao, Kaixuan Wang, Zhixuan Liang, Yusen Qin, Xiaokang Yang, Ping Luo, and Yao Mu. RoboTwin 2.0: A Scalable Data Generator and Benchmark with Strong Domain Randomization for Robust Bimanual Robotic Manipulation. *CoRR*, abs/2506.18088, 2025. URL https://arxiv.org/abs/2506.18088. 3

Tailai Cheng, Kejia Chen, Lingyun Chen, Liding Zhang, Yue Zhang, Yao Ling, Mahdi Hamad, Zhenshan Bing, Fan Wu, Karan Sharma, and Alois Knoll. TacUMI: A Multi-Modal Universal Manipulation Interface for Contact-Rich Tasks. *CoRR*, abs/2601.14550, 2026a. URL https://arxiv.org/abs/2601.14550. 5, 13

Zhengxue Cheng, Yiqian Zhang, Anni Tang, Keyu Wang, Wenkang Zhang, Haoyu Li, Hengdi Zhang, and Li Song. OmniVTLA: Vision-Tactile-Language-Action Models With Semantic-Aligned Tactile Sensing. *IEEE Robotics and Automation Letters*, 11(9):10146–10153, 2026b. doi: 10.1109/lra.2026.3710060. URL https://doi.org/10.1109/lra.2026.3710060. 5

Cheng Chi, Zhenjia Xu, Chuer Pan, Eric Cousineau, Benjamin Burchfiel, Siyuan Feng, Russ Tedrake, and Shuran Song. Universal Manipulation Interface: In-The-Wild Robot Teaching Without In-The-Wild Robots. In *Proceedings of Robotics: Science and Systems*, Delft, Netherlands, 2024. doi: 10.15607/rss.2024.xx.045. URL https://doi.org/10.15607/rss.2024.xx.045. 4, 11, 13

Hojung Choi, Yifan Hou, Chuer Pan, Seongheon Hong, Austin Patel, Xiaomeng Xu, Mark R. Cutkosky, and Shuran Song. In-the-Wild Compliant Manipulation with UMI-FT. *CoRR*, abs/2601.09988, 2026. URL https://arxiv.org/abs/2601.09988. 5, 8, 9

An Thanh Dang, Jayjun Lee, Mustafa Mukadam, X. Alice Wu, Bernadette Bucher, Manikantan Nambi, and Nima Fazeli. HydroShear: Hydroelastic Shear Simulation for Tactile Sim-to-Real Reinforcement Learning. In *Proceedings of Robotics: Science and Systems*, Sydney, Australia, 2026. doi: 10.15607/rss.2026.xxii.155. URL https://doi.org/10.15607/rss.2026.xxii.155. 4

Shengliang Deng, Mi Yan, Songlin Wei, Haixin Ma, Yuxin Yang, Jiayi Chen, Zhiqi Zhang, Taoyu Yang, Xuheng Zhang, Heming Cui, Zhizheng Zhang, and He Wang. GraspVLA: a Grasping Foundation Model Pre-trained on Billion-scale Synthetic Action Data. In Joseph Lim, Shuran Song, and Hae-Won Park, editors, *Proceedings of The 9th Conference on Robot Learning*, volume 305 of *Proceedings of Machine Learning Research*, pages 1004–1029. PMLR, 2025. URL https://proceedings.mlr.press/v305/deng25a.html. 3

Ruoxuan Feng, Jiangyu Hu, Wenke Xia, Tianci Gao, Ao Shen, Yuhao Sun, Bin Fang, and Di Hu. AnyTouch: Learning Unified Static-Dynamic Representation across Multiple Visuo-tactile Sensors. In Y. Yue, A. Garg, N. Peng, F. Sha, and R. Yu, editors, *International Conference on Learning Representations*, volume 2025, pages 31265–31285, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/file/4d893f766ab60e5337659b9e71883af4-Paper-Conference.pdf. 5

Ruoxuan Feng, Yuxuan Zhou, Siyu Mei, Dongzhan Zhou, Pengwei Wang, Shaowei Cui, Bin Fang, Guocai Yao, and Di Hu. AnyTouch 2: General Optical Tactile Representation Learning For Dynamic Tactile Perception. In C. Vondrick, B. Hariharan, C. Raffel, L. Pinto, D. Yang, and A. Faust, editors, *International Conference on Learning Representations*, volume 2026, pages 3543–3577, 2026. URL https://proceedings.iclr.cc/paper_files/paper/2026/file/073c8584ef86bee26fe9d639ec648e28-Paper-Conference.pdf. 5

Martin A. Fischler and Robert C. Bolles. Random sample consensus: a paradigm for model fitting with applications to image analysis and automated cartography. *Communications of the ACM*, 24(6):381–395, 1981. doi: 10.1145/358669.358692. URL https://doi.org/10.1145/358669.358692. 15

Kipp Freud, Yijiong Lin, and Nathan F. Lepora. SimShear: Sim-to-Real Shear-based Tactile Servoing. In Joseph Lim, Shuran Song, and Hae-Won Park, editors, *Proceedings of The 9th Conference on Robot Learning*, volume 305 of *Proceedings of Machine Learning Research*, pages 3401–3412. PMLR, 2025. URL https://proceedings.mlr.press/v305/freud25a.html. 4, 14

Caelan Reed Garrett, Ajay Mandlekar, Bowen Wen, and Dieter Fox. SkillMimicGen: Automated Demonstration Generation for Efficient Skill Learning and Deployment. In Pulkit Agrawal, Oliver Kroemer, and Wolfram Burgard, editors, *Proceedings of The 8th Conference on Robot Learning*, volume 270 of *Proceedings of Machine Learning Research*, pages 2750–2790. PMLR, 2025. URL https://proceedings.mlr.press/v270/garrett25a.html. 3

Haoran Geng, Feishi Wang, Songlin Wei, Yuyang Li, Bangjun Wang, Boshi An, Haozhe Lou, Charlie Tianyue Cheng, Peihao Li, Haozhe Chen, Yutong Liang, Yuxi Qian, Jiageng Mao, Weikang Wan, Yiran Geng, Mingtong Zhang, Jiangran Lyu, Siheng Zhao, Jiazhao Zhang, Chaoyi Xu, Jialiang Zhang, Chengyang Zhao, Haoran Lu, Yufei Ding, Ran Gong, Yuran Wang, Yuxuan Kuang, Ruihai Wu, Baoxiong Jia, Hao Dong, Siyuan Huang, Yue Wang, Jitendra Malik, and Pieter Abbeel. RoboVerse: A Unified Platform, Benchmark and Dataset for Scalable and Generalizable Robot Learning. In *Proceedings of Robotics: Science and Systems*, LosAngeles, CA, USA, 2025. doi: 10.15607/rss.2025.xxi.022. URL https://doi.org/10.15607/rss.2025.xxi.022. 3

Diandian Gu, Jing Lin, Gaohong Liu, Jiahang Liu, Su Ma, Guang Shi, Jun Wang, Qinlong Wang, Qianyi Wu, Zhongcong Xu, Xuanyu Yi, Zihao Yu, Jianfeng Zhang, Zhuolin Zheng, Yifan Zhu, Rui Chen, Hengkai Guo, Xiaoyang Guo, Mingcong Han, Xu Han, Xiu Li, Yixun Liang, Weiqiang Lou, Junzhe Lu, Guan Luo, Minghan Qin, Shuguang Wang, and Yuang Wang. Seed3D 2.0: Advancing High-Fidelity Simulation-Ready 3D Content Generation. *CoRR*, abs/2605.13862, 2026. URL https://arxiv.org/abs/2605.13862. 3, 16

Yanjiang Guo, Lucy Shi, Jianyu Chen, and Chelsea Finn. Ctrl-World: A Controllable Generative World Model for Robot Manipulation. In C. Vondrick, B. Hariharan, C. Raffel, L. Pinto, D. Yang, and A. Faust, editors, *International Conference on Learning Representations*, volume 2026, pages 6121–6138, 2026. URL https://proceedings.iclr.cc/paper_files/paper/2026/file/0ae94013da7cd459402fd77874e09ee3-Paper-Conference.pdf. 3, 23, 24

Harsh Gupta, Yuchen Mo, Shengmiao Jin, and Wenzhen Yuan. Sensor-Invariant Tactile Representation. In Y. Yue, A. Garg, N. Peng, F. Sha, and R. Yu, editors, *International Conference on Learning Representations*, volume 2025, pages 96515–96538, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/file/f05a11e682bf9c57ab62f7cff08fbe5b-Paper-Conference.pdf. 5

Peng Hao, Chaofan Zhang, Dingzhe Li, Xiaoge Cao, Xiaoshuai Hao, Shaowei Cui, and Shuo Wang. TLA: tactile-language-action model for contact-rich manipulation. *Robot Learning*, pages 1–1, 2026. doi: 10.55092/rl20260001. URL https://doi.org/10.55092/rl20260001. 5

Carolina Higuera, Akash Sharma, Chaithanya Krishna Bodduluri, Taosha Fan, Patrick Lancaster, Mrinal Kalakrishnan, Michael Kaess, Byron Boots, Mike Lambeta, Tingfan Wu, and Mustafa Mukadam. Sparsh: Self-supervised touch representations for vision-based tactile sensing. In Pulkit Agrawal, Oliver Kroemer, and Wolfram Burgard, editors, *Proceedings of The 8th Conference on Robot Learning*, volume 270 of *Proceedings of Machine Learning Research*, pages 885–915. PMLR, 2025a. URL https://proceedings.mlr.press/v270/higuera25a.html. 5

Carolina Higuera, Akash Sharma, Taosha Fan, Chaithanya Krishna Bodduluri, Byron Boots, Michael Kaess, Mike Lambeta, Tingfan Wu, Zixi Liu, Francois Robert Hogan, and Mustafa Mukadam. Tactile Beyond Pixels: Multisensory Touch Representations for Robot Manipulation. In Joseph Lim, Shuran Song, and Hae-Won Park, editors, *Proceedings of The 9th Conference on Robot Learning*, volume 305 of *Proceedings of Machine Learning Research*, pages 105–123. PMLR, 2025b. URL https://proceedings.mlr.press/v305/higuera25a.html. 5

Neville Hogan. Impedance Control: An Approach to Manipulation: Part I—Theory. *Journal of Dynamic Systems, Measurement, and Control*, 107(1):1–7, 1985. doi: 10.1115/1.3140702. URL https://doi.org/10.1115/1.3140702. 7, 9

HTC VIVE. VIVE Tracker (3.0), n.d. URL https://shop-us.vive.com/products/vive-tracker-3-0-full-body-tracking. Official product documentation. Accessed November 5, 2026. 13

Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-Rank Adaptation of Large Language Models. In *International Conference on Learning Representations*, 2022. URL https://openreview.net/forum?id=nZeVKeeFYf9. 20

Binghao Huang, Yixuan Wang, Xinyi Yang, Yiyue Luo, and Yunzhu Li. 3D-ViTac: Learning Fine-Grained Manipulation with Visuo-Tactile Sensing. In Pulkit Agrawal, Oliver Kroemer, and Wolfram Burgard, editors, *Proceedings of The 8th Conference on Robot Learning*, volume 270 of *Proceedings of Machine Learning Research*, pages 2557–2578. PMLR, 2025a. URL https://proceedings.mlr.press/v270/huang25e.html. 4

Binghao Huang, Jie Xu, Iretiayo Akinola, Wei Yang, Balakumar Sundaralingam, Rowland O'Flaherty, Dieter Fox, Xiaolong Wang, Arsalan Mousavian, Yu-Wei Chao, and Yunzhu Li. VT-Refine: Learning Bimanual Assembly with Visuo-Tactile Feedback via Simulation Fine-Tuning. In Joseph Lim, Shuran Song, and Hae-Won Park, editors, *Proceedings of The 9th Conference on Robot Learning*, volume 305 of *Proceedings of Machine Learning Research*, pages 2484–2500. PMLR, 2025b. URL https://proceedings.mlr.press/v305/huang25b.html. 4

Jialei Huang, Shuo Wang, Fanqi Lin, Yihang Hu, Chuan Wen, and Yang Gao. Tactile-VLA: Unlocking Vision-Language-Action Model's Physical Knowledge for Tactile Generalization. *CoRR*, abs/2507.09160, 2025c. URL https://arxiv.org/abs/2507.09160. 5, 7, 9, 20

Arhan Jain, Mingtong Zhang, Kanav Arora, William Chen, Marcel Torne, Muhammad Zubair Irshad, Sergey Zakharov, Yue Wang, Sergey Levine, Chelsea Finn, Wei-Chiu Ma, Dhruv Shah, Abhishek Gupta, and Karl Pertsch. PolaRiS: Scalable Real-to-Sim Evaluations for Generalist Robot Policies. In *Proceedings of Robotics: Science and Systems*, Sydney, Australia, 2026. doi: 10.15607/rss.2026.xxii.062. URL https://doi.org/10.15607/rss.2026.xxii.062. 3, 14, 22, 23, 24

Bowen Jing, Mingxin Wang, Ruiyang Hao, Chenchen Ge, Hanwen Shen, Junjie He, Yang Cui, Yiming Hou, Weitao Zhou, Jiawei Wang, Minglei Li, Dandan Zhang, Ding Zhao, Houde Liu, Xiaofan Li, Si Liu, Ping Luo, and Haibao Yu. SoftVTBench: A Deformation-Aware Visuo-Tactile Dataset and Benchmark for Deformable-Object Manipulation. *CoRR*, abs/2608.18701, 2026. URL https://arxiv.org/abs/2608.18701. 3, 7

Micah K. Johnson and Edward H. Adelson. Retrographic sensing for the measurement of surface texture and shape. In *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pages 1070–1077. IEEE, 2009. doi: 10.1109/cvpr.2009.5206534. URL https://doi.org/10.1109/cvpr.2009.5206534. 6

Bernhard Kerbl, Georgios Kopanas, Thomas Leimkuehler, and George Drettakis. 3D Gaussian Splatting for Real-Time Radiance Field Rendering. *ACM Transactions on Graphics*, 42(4):1–14, 2023. doi: 10.1145/3592433. URL https://doi.org/10.1145/3592433. 15

O. Khatib. A unified approach for motion and force control of robot manipulators: The operational space formulation. *IEEE Journal on Robotics and Automation*, 3(1):43–53, 1987. doi: 10.1109/jra.1987.1087068. URL https://doi.org/10.1109/jra.1987.1087068. 6

Alexander Khazatsky, Karl Pertsch, Suraj Nair, Ashwin Balakrishna, Sudeep Dasari, Siddharth Karamcheti, Soroush Nasiriany, Mohan Kumar Srirama, Lawrence Yunliang Chen, Kirsty Ellis, Peter David Fagan, Joey Hejna, Masha Itkina, Marion Lepert, Yecheng Jason Ma, Patrick Tree Miller, Jimmy Wu, Suneel Belkhale, Shivin Dass, Huy Ha, Arhan Jain, Abraham Lee, Youngwoon Lee, Marius Memmel, Sungjae Park, Ilija Radosavovic, Kaiyuan Wang, Albert Zhan, Kevin Black, Cheng Chi, Kyle Beltran Hatch, Shan Lin, Jingpei Lu, Jean Mercat, Abdul Rehman, Pannag R Sanketi, Archit Sharma, Cody Simpson, Quan Vuong, Homer Rich Walke, Blake Wulfe, Ted Xiao, Jonathan Heewon Yang, Arefeh Yavary, Tony Z. Zhao, Christopher Agia, Rohan Baijal, Mateo Guaman Castro, Daphne Chen, Qiuyu Chen, Trinity Chung, Jaimyn Drake, Ethan Paul Foster, Jensen Gao, David Antonio Herrera, Minho Heo, Kyle Hsu, Jiaheng Hu, Donovon Jackson, Charlotte Le, Yunshuang Li, Roy Lin, Zehan Ma, Abhiram Maddukuri, Suvir Mirchandani, Daniel Morton, Tony Nguyen, Abigail O'Neill, Rosario Scalise, Derick Seale, Victor Son, Stephen Tian, Emi Tran, Andrew E. Wang, Yilin Wu, Annie Xie, Jingyun Yang, Patrick Yin, Yunchu Zhang, Osbert Bastani, Glen Berseth, Jeannette Bohg, Ken Goldberg, Abhinav Gupta, Abhishek Gupta, Dinesh Jayaraman, Joseph J Lim, Jitendra Malik,

Roberto Martín-Martín, Subramanian Ramamoorthy, Dorsa Sadigh, Shuran Song, Jiajun Wu, Michael C. Yip, Yuke Zhu, Thomas Kollar, Sergey Levine, and Chelsea Finn. DROID: A Large-Scale In-The-Wild Robot Manipulation Dataset. In *Proceedings of Robotics: Science and Systems*, Delft, Netherlands, 2024. doi: 10.15607/rss.2024.xx.120. URL https://doi.org/10.15607/rss.2024.xx.120. 4

Moo Jin Kim, Chelsea Finn, and Percy Liang. Fine-Tuning Vision-Language-Action Models: Optimizing Speed and Success. In *Proceedings of Robotics: Science and Systems*, LosAngeles, CA, USA, 2025a. doi: 10.15607/rss.2025.xxi.017. URL https://doi.org/10.15607/rss.2025.xxi.017. 5

Moo Jin Kim, Karl Pertsch, Siddharth Karamcheti, Ted Xiao, Ashwin Balakrishna, Suraj Nair, Rafael Rafailov, Ethan P Foster, Pannag R Sanketi, Quan Vuong, Thomas Kollar, Benjamin Burchfiel, Russ Tedrake, Dorsa Sadigh, Sergey Levine, Percy Liang, and Chelsea Finn. OpenVLA: An Open-Source Vision-Language-Action Model. In Pulkit Agrawal, Oliver Kroemer, and Wolfram Burgard, editors, *Proceedings of The 8th Conference on Robot Learning*, volume 270 of *Proceedings of Machine Learning Research*, pages 2679–2713. PMLR, 2025b. URL https://proceedings.mlr.press/v270/kim25c.html. 5, 18

Dale A. Lawrence. Stability and transparency in bilateral teleoperation. *IEEE Transactions on Robotics and Automation*, 9(5):624–637, 1993. doi: 10.1109/70.258054. URL https://doi.org/10.1109/70.258054. 12

Luc Le Tien, Alin Albu-Schäffer, Alessandro De Luca, and Gerd Hirzinger. Friction observer and compensation for control of robots with joint torque measurement. In *2008 IEEE/RSJ International Conference on Intelligent Robots and Systems*, pages 3789–3795, 2008. URL https://elib.dlr.de/55742/. 13

Chuanyu Li, Chaoyi Liu, Daotan Wang, Shuyu Zhang, Lusong Li, Zecui Zeng, Fangchen Liu, Jing Xu, and Rui Chen. ViTaMIn-B: A Reliable and Efficient Visuo-Tactile Bimanual Manipulation Interface. *CoRR*, abs/2511.05858, 2025a. URL https://arxiv.org/abs/2511.05858. 5, 13, 34, 36

Xuanlin Li, Kyle Hsu, Jiayuan Gu, Oier Mees, Karl Pertsch, Homer Rich Walke, Chuyuan Fu, Ishikaa Lunawat, Isabel Sieh, Sean Kirmani, Sergey Levine, Jiajun Wu, Chelsea Finn, Hao Su, Quan Vuong, and Ted Xiao. Evaluating Real-World Robot Manipulation Policies in Simulation. In Pulkit Agrawal, Oliver Kroemer, and Wolfram Burgard, editors, *Proceedings of The 8th Conference on Robot Learning*, volume 270 of *Proceedings of Machine Learning Research*, pages 3705–3728. PMLR, 2025b. URL https://proceedings.mlr.press/v270/li25c.html. 3, 14, 23

Yichen Li, Yilun Du, Chao Liu, Francis Williams, Michael Foshey, Benjamin Eckart, Jan Kautz, Joshua B Tenenbaum, Antonio Torralba, and Wojciech Matusik. Learning to Jointly Understand Visual and Tactile Signals. In B. Kim, Y. Yue, S. Chaudhuri, K. Fragkiadaki, M. Khan, and Y. Sun, editors, *International Conference on Learning Representations*, volume 2024, pages 13153–13169, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/file/3848bc3112429079af85dedb7d369ef4-Paper-Conference.pdf. 5

Yuyang Li, Wenxin Du, Chang Yu, Puhao Li, Zihang Zhao, Tengyu Liu, Chenfanfu Jiang, Yixin Zhu, and Siyuan Huang. Taccel: Scaling Up Vision-based Tactile Robotics via High-performance GPU Simulation. In D. Belgrave, C. Zhang, H. Lin, R. Pascanu, P. Koniusz, M. Ghassemi, and N. Chen, editors, *Advances in Neural Information Processing Systems*, volume 38, Main Conference, pages 94577–94604. Curran Associates, Inc., 2025c. doi: 10.52202/085713-3162. URL https://doi.org/10.52202/085713-3162. 4

Bo Liu, Yifeng Zhu, Chongkai Gao, Yihao Feng, Qiang Liu, Yuke Zhu, and Peter Stone. LIBERO: Benchmarking Knowledge Transfer for Lifelong Robot Learning. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in Neural Information Processing Systems*, volume 36, pages 44776–44791. Curran Associates, Inc., 2023. doi: 10.52202/075280-1939. URL https://doi.org/10.52202/075280-1939. 3

Fangchen Liu, Chuanyu Li, Yihua Qin, Jing Xu, Pieter Abbeel, and Rui Chen. ViTaMIn: Learning Contact-Rich Tasks Through Robot-Free Visuo-Tactile Manipulation Interface. *CoRR*, abs/2504.06156, 2025a. URL https://arxiv.org/abs/2504.06156. 5

Qingtao Liu, Yu Cui, Zhengnan Sun, Gaofeng Li, Jiming Chen, and Qi Ye. VTDexManip: A Dataset and Benchmark for Visual-tactile Pretraining and Dexterous Manipulation with Reinforcement Learning. In Y. Yue, A. Garg, N. Peng, F. Sha, and R. Yu, editors, *International Conference on Learning Representations*, volume 2025, pages 90582–90607, 2025b. URL https://proceedings.iclr.cc/paper_files/paper/2025/file/e19b6f65791e350347bcff8a3955cb5b-Paper-Conference.pdf. 3

Songming Liu, Lingxuan Wu, Bangguo Li, Hengkai Tan, Huayu Chen, Zhengyi Wang, Ke Xu, Hang Su, and Jun Zhu. RDT-1B: a Diffusion Foundation Model for Bimanual Manipulation. In Y. Yue, A. Garg, N. Peng, F. Sha, and R. Yu, editors, *International Conference on Learning Representations*, volume 2025, pages 29982–30009, 2025c. URL https://proceedings.iclr.cc/paper_files/paper/2025/file/49f80e4d2471ad4f2edf4f5f1ab62339-Paper-Conference.pdf. 5

Rundong Luo, Yunong Liu, Di Cao, Matthew Tancik, Shyamal Buch, Colton Stearns, and Wenqi Xian. SceneRig: An Agentic System for Simulation-Ready 3D Scene Reconstruction from Single Images. *CoRR*, abs/2610.28585, 2026. URL https://arxiv.org/abs/2610.28585. 3, 23, 24

Oier Mees, Lukas Hermann, Erick Rosete-Beas, and Wolfram Burgard. CALVIN: A Benchmark for Language-Conditioned Policy Learning for Long-Horizon Robot Manipulation Tasks. *IEEE Robotics and Automation Letters*, 7(3):7327–7334, 2022. doi: 10.1109/lra.2022.3180108. URL https://doi.org/10.1109/lra.2022.3180108. 3

Yao Mu, Tianxing Chen, Zanxin Chen, Shijia Peng, Zhiqian Lan, Zeyu Gao, Zhixuan Liang, Qiaojun Yu, Yude Zou, Mingkun Xu, Lunkai Lin, Zhiqiang Xie, Mingyu Ding, and Ping Luo. RoboTwin: Dual-Arm Robot Benchmark with Generative Digital Twins. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 27649–27660, 2025. 3

Soroush Nasiriany, Abhiram Maddukuri, Lance Zhang, Adeet Parikh, Aaron Lo, Abhishek Joshi, Ajay Mandlekar, and Yuke Zhu. RoboCasa: Large-Scale Simulation of Household Tasks for Generalist Robots. In *Proceedings of Robotics: Science and Systems*, Delft, Netherlands, 2024. doi: 10.15607/rss.2024.xx.050. URL https://doi.org/10.15607/rss.2024.xx.050. 3

Michal Nazarczuk, Karla Stepanova, Jan Kristof Behrens, Matej Hoffmann, and Krystian Mikolajczyk. MuBlE: MuJoCo and Blender simulation Environment and Benchmark for Task Planning in Robot Manipulation. *CoRR*, abs/2503.02834, 2025. URL https://arxiv.org/abs/2503.02834. 3

Duc Huy Nguyen, Tim Schneider, Guillaume Duret, Alap Kshirsagar, Boris Belousov, and Jan Peters. TacEx: GelSight Tactile Simulation in Isaac Sim – Combining Soft-Body and Visuotactile Simulators. *CoRR*, abs/2411.04776, 2024. URL https://arxiv.org/abs/2411.04776. 3

NVIDIA, Johan Bjorck, Fernando Castañeda, Nikita Cherniadev, Xingye Da, Runyu Ding, Linxi "Jim" Fan, Yu Fang, Dieter Fox, Fengyuan Hu, Spencer Huang, Joel Jang, Zhenyu Jiang, Jan Kautz, Kaushil Kundalia, Lawrence Lao, Zhiqi Li, Zongyu Lin, Kevin Lin, Guilin Liu, Edith Llontop, Loic Magne, Ajay Mandlekar, Avnish Narayan, Soroush Nasiriany, Scott Reed, You Liang Tan, Guanzhi Wang, Zu Wang, Jing Wang, Qi Wang, Jiannan Xiang, Yuqi Xie, Yinzhen Xu, Zhenjia Xu, Seonghyeon Ye, Zhiding Yu, Ao Zhang, Hao Zhang, Yizhou Zhao, Ruijie Zheng, and Yuke Zhu. GR00T N1: An Open Foundation Model for Generalist Humanoid Robots. *CoRR*, abs/2503.14734, 2025a. URL https://arxiv.org/abs/2503.14734. 5

NVIDIA, Mayank Mittal, Pascal Roth, James Tigue, Antoine Richard, Octi Zhang, Peter Du, Antonio Serrano-Muñoz, Xinjie Yao, René Zurbrügg, Nikita Rudin, Lukasz Wawrzyniak, Milad Rakhsha, Alain Denzler, Eric

Heiden, Ales Borovicka, Ossama Ahmed, Iretiayo Akinola, Abrar Anwar, Mark T. Carlson, Ji Yuan Feng, Animesh Garg, Renato Gasoto, Lionel Gulich, Yijie Guo, M. Gussert, Alex Hansen, Mihir Kulkarni, Chenran Li, Wei Liu, Viktor Makoviychuk, Grzegorz Malczyk, Hammad Mazhar, Masoud Moghani, Adithyavairavan Murali, Michael Noseworthy, Alexander Poddubny, Nathan Ratliff, Welf Rehberg, Clemens Schwarke, Ritvik Singh, James Latham Smith, Bingjie Tang, Ruchik Thaker, Matthew Trepte, Karl Van Wyk, Fangzhou Yu, Alex Millane, Vikram Ramasamy, Remo Steiner, Sangeeta Subramanian, Clemens Volk, CY Chen, Neel Jawale, Ashwin Varghese Kuruttukulam, Michael A. Lin, Ajay Mandlekar, Karsten Patzwaldt, John Welsh, Huihua Zhao, Fatima Anes, Jean-Francois Lafleche, Nicolas Moënne-Loccoz, Soowan Park, Rob Stepinski, Dirk Van Gelder, Chris Amevor, Jan Carius, Jumyung Chang, Anka He Chen, Pablo de Heras Ciechomski, Gilles Daviet, Mohammad Mohajerani, Julia von Muralt, Viktor Reutskyy, Michael Sauter, Simon Schirm, Eric L. Shi, Pierre Terdiman, Kenny Vilella, Tobias Widmer, Gordon Yeoman, Tiffany Chen, Sergey Grizan, Cathy Li, Lotus Li, Connor Smith, Rafael Wiltz, Kostas Alexis, Yan Chang, David Chu, Linxi "Jim" Fan, Farbod Farshidian, Ankur Handa, Spencer Huang, Marco Hutter, Yashraj Narang, Soha Pouya, Shiwei Sheng, Yuke Zhu, Miles Macklin, Adam Moravanszky, Philipp Reist, Yunrong Guo, David Hoeller, and Gavriel State. Isaac Lab: A GPU-Accelerated Simulation Framework for Multi-Modal Robot Learning. *CoRR*, abs/2511.04831, 2025b. URL https://arxiv.org/abs/2511.04831. 3, 6

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning Transferable Visual Models From Natural Language Supervision. In *International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pages 8748–8763. PMLR, 2021. URL https://proceedings.mlr.press/v139/radford21a.html. 21

Nadun Ranawaka, Josiah Wong, Wei-Lin Pai, Wei-Teng Chu, Tianyuan Dai, Masoud Moghani, Hang Yin, Yunfan Jiang, Wesley Durbano, Brandon Huynh, Yu Fang, Danfei Xu, Ruohan Zhang, Li Fei-Fei, Linxi Fan, Bowen Wen, Ajay Mandlekar, and Yuke Zhu. SimFoundry: Modular and Automated Scene Generation for Policy Learning and Evaluation. *CoRR*, abs/2606.28276, 2026. URL https://arxiv.org/abs/2606.28276. 3, 6, 23, 24

Samanta Rodriguez, Yiming Dou, Miquel Oller, Andrew Owens, and Nima Fazeli. Cross-Sensor Touch Generation. In Joseph Lim, Shuran Song, and Hae-Won Park, editors, *Proceedings of The 9th Conference on Robot Learning*, volume 305 of *Proceedings of Machine Learning Research*, pages 152–167. PMLR, 2025. URL https://proceedings.mlr.press/v305/rodriguez25a.html. 4

Johannes L. Schonberger and Jan-Michael Frahm. Structure-From-Motion Revisited. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2016. 15

Zilin Si and Wenzhen Yuan. Taxim: An Example-Based Simulation Model for GelSight Tactile Sensors. *IEEE Robotics and Automation Letters*, 7(2):2361–2368, 2022. doi: 10.1109/lra.2022.3142412. URL https://doi.org/10.1109/lra.2022.3142412. 3, 6, 16, 34

Zilin Si, Gu Zhang, Qingwei Ben, Branden Romero, Zhou Xian, Chao Liu, and Chuang Gan. DIFFTACTILE: A Physics-based Differentiable Tactile Simulator for Contact-rich Robotic Manipulation. In B. Kim, Y. Yue, S. Chaudhuri, K. Fragkiadaki, M. Khan, and Y. Sun, editors, *International Conference on Learning Representations*, volume 2024, pages 7164–7183, 2024. URL https://proceedings.iclr.cc/paper_files/paper/2024/file/1cc70be9fb6a83bc46cf4ac21a91e0b0-Paper-Conference.pdf. 4

Stone Tao, Fanbo Xiang, Arth Shukla, Yuzhe Qin, Xander Hinrichsen, Xiaodi Yuan, Chen Bao, Xinsong Lin, Yulin Liu, Tse-Kai Chan, Yuan Gao, Xuanlin Li, Tongzhou Mu, Nan Xiao, Arnav Gurha, Viswesh Nagaswamy Rajesh, Yong Woo Choi, Yen-Ru Chen, Zhiao Huang, Roberto Calandra, Rui Chen, Shan Luo, and Hao Su. Demonstrating GPU Parallelized Robot Simulation and Rendering for Generalizable Embodied AI with

ManiSkill3. In *Proceedings of Robotics: Science and Systems*, LosAngeles, CA, USA, 2025. doi: 10.15607/rss.2025.xxi.021. URL https://doi.org/10.15607/rss.2025.xxi.021. 3

Shaoxiong Wang, Mike Lambeta, Po-Wei Chou, and Roberto Calandra. TACTO: A Fast, Flexible, and Open-Source Simulator for High-Resolution Vision-Based Tactile Sensors. *IEEE Robotics and Automation Letters*, 7(2): 3930–3937, 2022. doi: 10.1109/lra.2022.3146945. URL https://doi.org/10.1109/lra.2022.3146945. 3

Lai Wei, Jiahua Ma, Yibo Hu, and Ruimao Zhang. Ensuring Force Safety in Vision-Guided Robotic Manipulation via Implicit Tactile Calibration. In Joseph Lim, Shuran Song, and Hae-Won Park, editors, *Proceedings of The 9th Conference on Robot Learning*, volume 305 of *Proceedings of Machine Learning Research*, pages 1049–1062. PMLR, 2025. URL https://proceedings.mlr.press/v305/wei25a.html. 3, 7

Youngsun Wi, Jessica Yin, Elvis Xiang, Akash Sharma, Jitendra Malik, Mustafa Mukadam, Nima Fazeli, and Tess Hellebrekers. TactAlign: Human-to-Robot Policy Transfer via Tactile Alignment. In *Proceedings of Robotics: Science and Systems*, Sydney, Australia, 2026. doi: 10.15607/rss.2026.xxii.006. URL https://doi.org/10.15607/rss.2026.xxii.006. 14

Longyan Wu, Checheng Yu, Jieji Ren, Li Chen, Yufei Jiang, Ran Huang, Guoying Gu, and Hongyang Li. FreeTacMan: Robot-free Visuo-Tactile Data Collection System for Contact-rich Manipulation. *CoRR*, abs/2506.01941, 2025a. URL https://arxiv.org/abs/2506.01941. 5

Mingdong Wu, Long Yang, Jin Liu, Weiyao Huang, Lehong Wu, Zelin Chen, Daolin Ma, and Hao Dong. UniTac2Pose: A Unified Approach Learned in Simulation for Category-level Visuotactile In-hand Pose Estimation. In Joseph Lim, Shuran Song, and Hae-Won Park, editors, *Proceedings of The 9th Conference on Robot Learning*, volume 305 of *Proceedings of Machine Learning Research*, pages 4367–4384. PMLR, 2025b. URL https://proceedings.mlr.press/v305/wu25d.html. 4

Yifan Wu, Qin Li, Nan Min, Guojin Zhong, Haoyu Zhao, Zhiyuan Li, Houze Xu, Shengqi Xu, Xingyao Lin, Zijie Diao, Zhaoxiang Liu, Shiguo Lian, Shunlin Lu, Shihao Zhao, Ziyi Ye, Zuxuan Wu, and Yu-Gang Jiang. OpenViTac: Learning and Benchmarking Visuo-Tactile Policies in a Unified Sim-and-Real Framework. *CoRR*, abs/2610.10384, 2026. URL https://arxiv.org/abs/2610.10384. 3, 23, 24

Mengda Xu, Han Zhang, Yifan Hou, Zhenjia Xu, Linxi Fan, Manuela Veloso, and Shuran Song. DexUMI: Using Human Hand as the Universal Manipulation Interface for Dexterous Manipulation. In Joseph Lim, Shuran Song, and Hae-Won Park, editors, *Proceedings of The 9th Conference on Robot Learning*, volume 305 of *Proceedings of Machine Learning Research*, pages 437–459. PMLR, 2025a. URL https://proceedings.mlr.press/v305/xu25b.html. 14

Yue Xu, Litao Wei, Pengyu An, Qingyu Zhang, and Yong-Lu Li. exUMI: Extensible Robot Teaching System with Action-aware Task-agnostic Tactile Representation. In Joseph Lim, Shuran Song, and Hae-Won Park, editors, *Proceedings of The 9th Conference on Robot Learning*, volume 305 of *Proceedings of Machine Learning Research*, pages 2536–2554. PMLR, 2025b. URL https://proceedings.mlr.press/v305/xu25e.html. 5

Han Xue, Jieji Ren, Wendi Chen, Gu Zhang, Fang Yuan, Guoying Gu, Huazhe Xu, and Cewu Lu. Reactive Diffusion Policy: Slow-Fast Visual-Tactile Policy Learning for Contact-Rich Manipulation. In *Proceedings of Robotics: Science and Systems*, LosAngeles, CA, USA, 2025. doi: 10.15607/rss.2025.xxi.052. URL https://doi.org/10.15607/rss.2025.xxi.052. 5, 11

Fengyu Yang, Chao Feng, Ziyang Chen, Hyoungseob Park, Daniel Wang, Yiming Dou, Ziyao Zeng, Xien Chen, Rit Gangopadhyay, Andrew Owens, and Alex Wong. Binding Touch to Everything: Learning Unified Multimodal Tactile Representations. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 26340–26353, 2024. 5

Max Yang, chenghua lu, Alex Church, Yijiong Lin, Christopher J. Ford, Haoran Li, Efi Psomopoulou, David A.W. Barton, and Nathan F. Lepora. AnyRotate: Gravity-Invariant In-Hand Object Rotation with Sim-to-Real Touch. In Pulkit Agrawal, Oliver Kroemer, and Wolfram Burgard, editors, *Proceedings of The 8th Conference on Robot Learning*, volume 270 of *Proceedings of Machine Learning Research*, pages 4727–4747. PMLR, 2025. URL https://proceedings.mlr.press/v270/yang25c.html. 4, 14

Kelin Yu, Yunhai Han, Qixian Wang, Vaibhav Saxena, Danfei Xu, and Ye Zhao. MimicTouch: Leveraging Multi-modal Human Tactile Demonstrations for Contact-rich Manipulation. In Pulkit Agrawal, Oliver Kroemer, and Wolfram Burgard, editors, *Proceedings of The 8th Conference on Robot Learning*, volume 270 of *Proceedings of Machine Learning Research*, pages 4844–4865. PMLR, 2025a. URL https://proceedings.mlr.press/v270/yu25c.html. 14

Samson Yu, Lin Kelvin, Anxing Xiao, Jiafei Duan, and Harold Soh. Octopi: Object Property Reasoning with Large Tactile-Language Models. In *Proceedings of Robotics: Science and Systems*, Delft, Netherlands, 2024. doi: 10.15607/rss.2024.xx.066. URL https://doi.org/10.15607/rss.2024.xx.066. 5

Samson Yu, Lin Kelvin, and Harold Soh. Demonstrating the Octopi-1.5 Visual-Tactile-Language Model. In *Proceedings of Robotics: Science and Systems*, LosAngeles, CA, USA, 2025b. doi: 10.15607/rss.2025.xxi.058. URL https://doi.org/10.15607/rss.2025.xxi.058. 5

Wenzhen Yuan, Siyuan Dong, and Edward Adelson. GelSight: High-Resolution Robot Tactile Sensors for Estimating Geometry and Force. *Sensors*, 17(12):2762, 2017. doi: 10.3390/s17122762. URL https://doi.org/10.3390/s17122762. 4, 6

Chaofan Zhang, Peng Hao, Xiaoge Cao, Xiaoshuai Hao, Shaowei Cui, and Shuo Wang. VTLA: Vision-Tactile-Language-Action model with preference learning for insertion manipulation. *Biomimetic Intelligence and Robotics*, page 100333, 2026. doi: 10.1016/j.birob.2026.100333. URL https://doi.org/10.1016/j.birob.2026.100333. 5, 11, 20

Di Zhang, Chengbo Yuan, Chuan Wen, Hai Zhang, Junqiao Zhao, and Yang Gao. KineDex: Learning Tactile-Informed Visuomotor Policies via Kinesthetic Teaching for Dexterous Manipulation. In Joseph Lim, Shuran Song, and Hae-Won Park, editors, *Proceedings of The 9th Conference on Robot Learning*, volume 305 of *Proceedings of Machine Learning Research*, pages 4123–4138. PMLR, 2025a. URL https://proceedings.mlr.press/v305/zhang25l.html. 14

Zongzheng Zhang, Haobo Xu, Zhuo Yang, Chenghao Yue, Zehao Lin, Huan-ang Gao, Ziwei Wang, and Hao Zhao. Elucidating the Design Space of Torque-aware Vision-Language-Action Models. In Joseph Lim, Shuran Song, and Hae-Won Park, editors, *Proceedings of The 9th Conference on Robot Learning*, volume 305 of *Proceedings of Machine Learning Research*, pages 4019–4037. PMLR, 2025b. URL https://proceedings.mlr.press/v305/zhang25k.html. 5, 20

Jialiang Zhao, Naveen Kuppuswamy, Siyuan Feng, Benjamin Burchfiel, and Edward Adelson. PolyTouch: A Robust Multi-Modal Tactile Sensor for Contact-Rich Manipulation Using Tactile-Diffusion Policies. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pages 104–110. IEEE, 2025a. doi: 10.1109/icra55743.2025.11128816. URL https://doi.org/10.1109/icra55743.2025.11128816. 4

Jialiang Zhao, Yuxiang Ma, Lirui Wang, and Edward Adelson. Transferable Tactile Transformers for Representation Learning Across Diverse Sensors and Tasks. In Pulkit Agrawal, Oliver Kroemer, and Wolfram Burgard, editors, *Proceedings of The 8th Conference on Robot Learning*, volume 270 of *Proceedings of Machine Learning Research*, pages 3766–3779. PMLR, 2025b. URL https://proceedings.mlr.press/v270/zhao25c.html. 5

Yongqiang Zhao, Kun Qian, Boyi Duan, and Shan Luo. FOTS: A Fast Optical Tactile Simulator for Sim2Real Learning of Tactile-Motor Robot Manipulation Skills. *IEEE Robotics and Automation Letters*, 9(6):5647–5654, 2024. doi: 10.1109/lra.2024.3396665. URL https://doi.org/10.1109/lra.2024.3396665. 3, 6, 16, 34

Zhaxizhuom Zhaxizhuoma, Kehui Liu, Chuyue Guan, Zhongjie Jia, Ziniu Wu, Xin Liu, Tianyu Wang, Shuai Liang, Pengan CHEN, Pingrui Zhang, Haoming Song, Delin Qu, Dong Wang, Zhigang Wang, Nieqing Cao, Yan Ding, Bin Zhao, and Xuelong Li. FastUMI: A Scalable and Hardware-Independent Universal Manipulation Interface with Dataset. In Joseph Lim, Shuran Song, and Hae-Won Park, editors, *Proceedings of The 9th Conference on Robot Learning*, volume 305 of *Proceedings of Machine Learning Research*, pages 3069–3093. PMLR, 2025. URL https://proceedings.mlr.press/v305/zhaxizhuoma25a.html. 4

Jiayu Zhou, Qiwei Wu, Jian Li, Zhe Chen, Xiaogang Xiong, and Renjing Xu. Gentle Manipulation Policy Learning via Demonstrations from VLM Planned Atomic Skills. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 40, pages 18855–18863, 2026. doi: 10.1609/aaai.v40i22.38955. URL https://doi.org/10.1609/aaai.v40i22.38955. 8

Yuke Zhu, Josiah Wong, Ajay Mandlekar, Roberto Martín-Martín, Abhishek Joshi, Kevin Lin, Abhiram Maddukuri, Soroush Nasiriany, and Yifeng Zhu. robosuite: A Modular Simulation Framework and Benchmark for Robot Learning. *CoRR*, abs/2009.12293, 2020. URL https://arxiv.org/abs/2009.12293. 3